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[Intervention Review]

Repositioning for pressure injury prevention in adults

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ABSTRACT

Rationale

Pressure injuries (or pressure ulcers) are localised damage to skin or tissue (or both) occurring over bony prominences, resulting from prolonged pressure or shear forces (or both). Adults receiving care in any healthcare setting can develop pressure injuries. Immobility, malnutrition, and reduced sensation are some known risk factors. Regular repositioning is a theoretically sound prevention strategy that is part of standard patient care.

This is the second update of a review published in 2014 and updated in 2020.

Objectives

To evaluate the benefits and harms, and cost-effectiveness, of repositioning regimens (i.e. repositioning frequencies, position, micromovement) for pressure injury prevention in adults in acute, long-term, or aged healthcare settings, compared to standard care or another repositioning regimen.

Search methods

To identify studies for inclusion in the review, we searched the Cochrane Central Register of Controlled Trials, Ovid MEDLINE, Embase, EBSCO CINAHL Plus, and trial registries on 7 May 2025. We also scanned the reference lists of included studies, reviews, meta-analyses, and health technology reports.

Eligibility criteria

We included randomised controlled trials that assessed the effects of any repositioning regimen and measured pressure injury incidence in adults (at least 18 years of age) without an existing pressure injury, in any acute, long-term, or aged healthcare setting.

Outcomes

The primary outcome was the cumulative incidence of pressure injuries (any category/stage). Secondary outcomes were health-related quality of life, procedural pain, patient satisfaction, pressure injury prevention and treatment costs, and incremental costs per pressure injury avoided.

Risk of bias

We assessed the risk of bias in the evidence using the Cochrane RoB 2 tool. We evaluated the certainty of the evidence using GRADE methodology and GRADEpro software.

Synthesis methods

Seven review authors were involved in independently undertaking study selection, data extraction, and RoB and GRADE assessment. All outcomes were binary and reported as risk ratios (RR) with 95% confidence intervals (CI). We pooled data using the fixed-effect or random-effects model, depending on clinical and methodology heterogeneity.

Included studies

We found three new trials. This 2026 update of the review therefore includes 11 trials, which were conducted in acute and aged healthcare settings, and involved 4462 participants aged 18 to 90 years. No new economic substudies were identified, so our cost analysis is based on the two economic evaluations that were included in the previous version of the review.

Synthesis of results

Primary outcome: incidence of new pressure injury (any category/stage)

Repositioning frequencies

Five trials evaluated different repositioning frequencies in intensive care unit (ICU) and nursing home settings. We pooled the results of four trials that compared 2- versus 4-hourly repositioning (RR 1.05, 95% CI 0.79 to 1.39; 4 trials, 1104 participants), but we judged the certainty of the evidence as very low. There were three other comparisons of repositioning frequencies: 2-hourly versus 3-hourly (RR 1.10, 95% CI 0.30 to 4.08; 3 trials, 795 participants); 3-hourly versus 4-hourly (RR 0.99, 95% CI 0.22 to 4.43; 3 trials, 776 participants); and 4-hourly versus 6-hourly (RR 0.73, 95% CI 0.53 to 1.02; 1 trial, 129 participants). We judged the certainty of the evidence for these results as very low. Therefore, for comparisons of different repositioning frequencies, the evidence is very uncertain.

One trial evaluated real-time wearable patient sensor data with visual reminders for 2-hourly repositioning compared to standard care (real-time patient sensor data without visual reminders and with nurse-initiated repositioning) in 1226 ICU patients. The study reported a significant reduction in pressure injuries with the visual warnings from the sensors (RR 0.28, 95% CI 0.10 to 0.75). We judged the certainty of this evidence as moderate.

Position

Four trials evaluated patient position. Two of the trials compared 30° tilt (and 3-hourly repositioning overnight) versus 90° tilt (and 6-hourly repositioning overnight) in acute and nursing home patients. We pooled their data and found an RR of 0.62 (95% CI 0.10 to 3.97; 2 trials, 252 participants), but we judged the certainty of this evidence as very low. One trial (120 participants) compared 30° and 45° tilts with 'usual care' in three ICUs and reported no pressure injuries in either trial arm, but we judged the certainty of this evidence as very low. One trial (116 participants) compared prone versus supine positioning in ICU patients, and found there may be higher pressure injury incidence in the prone group (RR 4.55, 95% CI 2.31 to 8.98). We judged the certainty of this evidence as low.

Micromovement

We pooled two trials comparing micromovement with standard care (silicone foam dressing) in operating rooms. The results suggested a reduction in pressure injuries with micromovement (RR 0.28, 95% CI 0.11 to 0.67; 2 trials, 477 participants), but the number of events was low. We judged the certainty of the evidence as low.

Secondary outcomes

No trials reported health-related quality of life, procedural pain, or patient satisfaction.

Cost analysis

Two trials, both conducted in nursing homes, included parallel economic evaluations. One compared costs of 2-hourly repositioning (321 participants) versus 3-hourly (326 participants) and 4-hourly (295 participants) schedules, with costs being \$11.05 and \$16.74 (Canadian dollars) less per resident per day for the 3- and 4-hourly regimens, respectively. The other compared 3-hourly repositioning using a 30° tilt versus standard care (6-hourly with a 90° lateral rotation) in 213 participants. It found that the intervention was cost-effective (EUR 206.60

versus 253.10, incremental difference EUR 46.50, 95% CI 1.25 to 74.60), with a projected annual saving of EUR 512,800 (equating to 21,462 nursing-care hours).

Authors' conclusions

Repositioning is a frequently used strategy for pressure injury prevention in adult patients in acute, long-term, and aged healthcare settings. This updated review includes three new trials, but the findings and conclusions align with our earlier reviews. Most of the evidence is of low or very low certainty. There is a lack of robust evaluation of repositioning regimens for pressure injury prevention, and studies are small, resulting in uncertainty about the review findings. There are limited economic evaluation data, making it difficult to reach reliable conclusions about the relative costs of different repositioning regimens.

Funding

None

Registration

Protocol (2012) DOI: 10.1002/14651858.CD009958

PLAIN LANGUAGE SUMMARY

How often should a patient's position be changed and which positions are best to prevent bedsores?

Key messages

- There is no clear evidence showing that any particular frequency of changing a person's position (repositioning) or any particular position (e.g. tilting the bed to a 30-degree angle) works better than any other to prevent bedsores (pressure injuries) in adults in hospital or care homes.
- Some repositioning frequencies and positions may reduce the costs of care, but this evidence is limited and uncertain.
- This is the second update of a review published in 2014. Research is still limited. Most studies are small and not well designed.

What are bedsores?

Bedsores (also called pressure injuries, pressure ulcers, pressure sores, decubitus ulcers) happen when people are not able to move much, or spend a lot of time sitting or lying down, such as elderly people or those who are very ill. When a body part presses against a mattress or chair for a long time, the skin rubs over bony parts of the body, like the heels, tailbone, hips, back, and the back of the head. Blood flow is reduced and skin and tissue break down.

How are bedsores prevented?

Changing a person's position can improve blood flow to areas that press against the bed or other surfaces, and it can have other benefits, such as increased comfort and mental well-being. The approaches used to prevent bedsores include:

- repositioning: moving or turning someone regularly, for example, every 2, 3, or 4 hours;
- position: placing someone in different positions, for example, raising the head of the bed or tilting the body;
- micromovement: frequent, small position changes made for people who cannot move (e.g. during surgery); and
- specially adapted mattresses and dressings.

What did we want to find out?

We wanted to know:

- which repositioning approaches (e.g. frequencies (how regularly someone is moved) and positions) are best for preventing bedsores;
- their effects on healthcare costs; and
- their effects on health-related quality of life, pain, and patient satisfaction.

What did we do?

We searched for studies that involved adults without bedsores who were receiving care in any hospital, long-term, or aged healthcare facility. The studies investigated which approaches to repositioning were most effective for preventing bedsores.

We summarised the findings and assessed our confidence in the results, based on how the studies were designed and conducted.

What did we find?

We found three new studies published since the 2020 version of the review. This 2026 updated review includes 11 studies and two cost evaluations.

The studies took place in China, Belgium, North America, Iran, and the UK. Most studies were in hospitals (intensive care units, operating theatres, wards); three were in nursing homes. The studies involved 4462 adults aged 18 to 90 years.

Repositioning

- We combined results from four studies comparing different repositioning frequencies. It is unclear whether changing a patient's position every 2 versus 4 hours, every 2 versus 3 hours, every 3 versus 4 hours, or every 4 versus 6 hours makes any difference to the chance of developing bedsores (4 studies, 2175 people).
- One study tested a device worn by patients that electronically tracks their position and feeds that information back to nurses. Nurses in one group got visual reminders from the sensor to turn the patient every two hours, while nurses in the other group did not get reminders and turned patients based on their own judgement. The visual warnings provided by the device probably reduce the chance of getting bedsores (1 study, 1312 people).

Position

- The effect of 30° tilt, 3-hourly repositioning overnight compared to 90° tilt, 6-hourly repositioning overnight on developing bedsores is unclear (2 studies, 252 people).
- Raising the head of the bed by 30° compared to 45° may have no effect on bedsores (1 study, 80 people), but the evidence is unclear.
- Lying face down (prone) with 'lung recruitment manoeuvres' (a procedure to help open up the lungs) may result in more bedsores than lying face up (1 study, 116 people).

Micromovement

- Micromovement may reduce bedsores (2 studies, 477 people).

Cost-effectiveness

- One study estimated that repositioning costs are lower per resident per day with 4-hourly compared to 2-hourly repositioning, mostly due to reduced nursing time.
- Another study estimated that repositioning every 3 hours with a 30-degree tilt costs less than standard care (repositioning every 6 hours with a 90-degree tilt).

Other measures

- No studies reported the other measures we were interested in.

What are the limitations of the evidence?

Our confidence in the evidence is low or very low. Most studies were small or poorly designed. Information about cost-effectiveness was very limited. Health-related quality of life, pain, and patient satisfaction were not reported at all.

As the evidence does not show the best way to prevent pressure injuries, how often to change someone's position should be based on the person's medical condition, personal preferences, comfort, and how much they can move themselves.

How up to date is this review?

This is an update of a review published in 2014 and updated in 2020. The evidence is current to 7 August 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - 2-hourly repositioning compared to 4-hourly repositioning for pressure injury prevention in adult patients

2-hourly repositioning compared to 4-hourly repositioning for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: intensive care unit and nursing homes

Intervention: 2-hourly repositioning

Comparison: 4-hourly repositioning

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with 4-hourly repositioning	Risk with 2-hourly repositioning				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	114 per 1000	120 per 1000 (90 to 159)	RR 1.05 (0.79 to 1.39)	1104 (4 RCTs)	⊕⊕⊕⊕ Very low ^{a,b}	
Health-related quality of life - not measured				-	-	
Procedural pain - not measured				-	-	
Patient satisfaction - not measured				-	-	
Costs (Costs)				616 (1 RCT)	-	Paulden (2014) reported repositioning costs were estimated to be lower per resident per day for the 4-hourly compared to the 2-hourly regimen, mostly due to reduced nursing time.
Incremental cost - not measured				-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_459938126520685400.

^a Four included studies: 1) Bergstrom 2013: overall ROB assessed as having 'Some concerns' in one domain for this result (Domain 2: deviations from intended intervention); 2) Defloor 2005: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results); 3) Manzano 2014: overall ROB assessed as having 'Some concerns' in one domain for this result (Domain 5: selection of reported results); and 4) Tarigan 2021: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results).

^b Downgraded one level for study limitations (risk of bias) as two studies had 'high' ROB and two studies had 'some concerns'. Downgraded one level for imprecision due to wide confidence intervals, which include the possibility of harm as well as benefit and no effect. Downgraded one level for a lack of blinding of nursing staff.

Summary of findings 2. Summary of findings table - 2-hourly repositioning compared to 3-hourly repositioning for pressure injury prevention in adult patients

2-hourly repositioning compared to 3-hourly repositioning for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: intensive care unit and nursing homes

Intervention: 2-hourly repositioning

Comparison: 3-hourly repositioning

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with 3-hourly repositioning	Risk with 2-hourly repositioning				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	113 per 1000	124 per 1000 (34 to 461)	RR 1.10 (0.30 to 4.08)	795 (3 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,c}	
Health-related quality of life - not measured				-	-	

Procedural pain - not measured		-	-	
Patient satisfaction - not measured		-	-	
Costs (Costs)		647 (1 RCT)	-	Paulden (2014) reported repositioning costs were estimated to be lower per resident per day for the 3-hourly compared to the 2-hourly regimen, mostly due to reduced nursing time.
Incremental costs - not measured		-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

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Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_460690209072194672.

^a Three included studies: 1) Bergstrom 2013: overall ROB assessed as having 'Some concerns' in one domain for this result (Domain 2: deviations from intended intervention); 2) Defloor 2005: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results); 3) Tarigan 2021: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results). Downgraded one level for study limitations (risk of bias) as two studies had 'high' ROB and one study had 'some concerns'.

^b Downgraded one level for significant heterogeneity (I² was 75%).

^c Downgraded one level for imprecision due to wide confidence intervals, which include the possibility of harm as well as benefit and no effect.

Summary of findings 3. Summary of findings table - 3-hourly repositioning compared to 4-hourly repositioning for pressure injury prevention in adult patients

3-hourly repositioning compared to 4-hourly repositioning for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: intensive care unit and nursing homes
Intervention: 3-hourly repositioning
Comparison: 4-hourly repositioning

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with 4-hourly repositioning	Risk with 3-hourly repositioning				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	106 per 1000	105 per 1000 (23 to 469)	RR 0.99 (0.22 to 4.43)	776 (3 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,c}	
Health-related quality of life - not measured				-	-	
Procedural pain - not measured				-	-	
Patient satisfaction - not measured				-	-	
Costs - not measured				-	-	
Incremental costs - not measured				-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_460690378832979224.

^a Three included studies: 1) Bergstrom 2013: overall ROB assessed as having 'Some concerns' in one domain for this result (Domain 2: deviations from intended intervention); 2) Defloor 2005: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results); 3) Tarigan 2021: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results). Downgraded one level for study limitations (risk of bias) as two studies had 'high' ROB and one study had 'some concerns'.

^b Downgraded one level for significant heterogeneity (I² was 75%).

^c Downgraded one level for imprecision due to wide confidence intervals which include the possibility of harm as well as benefit and no effect.

Summary of findings 4. Summary of findings table - 4-hourly repositioning compared to 6-hourly repositioning for pressure injury prevention in adult patients

4-hourly repositioning compared to 6-hourly repositioning for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: nursing homes

Intervention: 4-hourly repositioning

Comparison: 6-hourly repositioning

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with 6-hourly repositioning	Risk with 4-hourly repositioning				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	619 per 1000	452 per 1000 (328 to 631)	RR 0.73 (0.53 to 1.02)	129 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	
Health-related quality of life - not measured				-	-	
Procedural pain - not measured				-	-	
Patient satisfaction - not measured				-	-	
Costs - not measured				-	-	
Incremental cost - not measured				-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_460690414396220710.

^a One included study: Defloor, 2005: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results).

^b Downgraded one level for study limitations (risk of bias) as the study had 'high' ROB. Downgraded one level for imprecision due to wide confidence intervals which include the possibility of harm as well as benefit and no effect. Downgraded one level for a lack of blinding of nursing staff.

Summary of findings 5. Summary of findings table - Real-time patient sensor data with reminders for 2-hour repositioning compared to real-time patient sensor data without reminders and nurse-initiated repositioning for pressure injury prevention in adult patients

Real-time patient sensor data with reminders for 2-hour repositioning compared to real-time patient sensor data without reminders and nurse-initiated repositioning for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: intensive care unit

Intervention: real-time patient sensor data with reminders for 2-hour repositioning

Comparison: real-time patient sensor data without reminders and nurse-initiated repositioning

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with real-time patient sensor data without reminders and nurse-initiated repositioning	Risk with real-time patient sensor data with reminders for 2-hour repositioning				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	27 per 1000	8 per 1000 (3 to 20)	RR 0.28 (0.10 to 0.75)	1226 (1 RCT)	⊕⊕⊕⊖ Moderate ^{a,b}	
Health-related quality of life - not measured				-	-	
Procedural pain - not measured				-	-	
Patient satisfaction - not measured				-	-	
Costs - not measured				-	-	
Incremental costs - not measured				-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

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See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_460555343792240488.

^a One included study: Pickham, 2018: overall ROB assessed as having 'Some concerns' in one domain for this result (Domain 5: selection of reported results). ROB not downgraded as this did not impact the outcome.

^b Downgraded one level for imprecision due to small number of events.

Summary of findings 6. Summary of findings table - 30° tilt and 3-hourly repositioning overnight compared to 90° tilt and 6-hourly repositioning overnight for pressure injury prevention in adult patients

30° tilt and 3-hourly repositioning overnight compared to 90° tilt and 6-hourly repositioning overnight for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: acute and intensive care units

Intervention: 30° tilt and 3-hourly repositioning overnight

Comparison: 90° tilt and 6-hourly repositioning overnight

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with 90° tilt and 6-hourly repositioning overnight	Risk with 30° tilt and 3-hourly repositioning overnight				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	111 per 1000	69 per 1000 (11 to 441)	RR 0.62 (0.10 to 3.97)	252 (2 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,c}	
Health-related quality of life - not measured				-	-	

Procedural pain - not measured		-	-	
Patient satisfaction - not measured		-	-	
Costs (Costs)		213 (1 RCT)	-	Comparing the nursing time costs and the pressure injury incidence over a four-week period, Moore (2013) estimated that implementing a 30° tilt and 3-hourly regimen (intervention) instead of a 90° tilt and 6-hourly regimen (standard care) would be associated with an annual cost difference favouring the intervention of EUR 512,800, equivalent to 21,462 hours of nurse time.
Incremental cost (Incremental cost)		213 (1 RCT)	-	The intervention was reported to be cost-effective (EUR 206.60 versus 253.10, incremental difference EUR 46.50, 95% CI 1.25 to 74.60).

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_459938143565550428.

^a Two included studies: 1) Young 2004: overall ROB assessed as having 'Some concerns' in one domain for this result (Domain 2: deviations from intended intervention); 2) Moore 2011: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 1: randomisation; Domain 4: measuring the outcome; and Domain 5: selection of reported results). Downgraded one level for study limitations (risk of bias) as one study had 'high' ROB and one study had 'some concerns'.

^b Downgraded one level for significant heterogeneity (I² was 69%).

^c Downgraded one level for imprecision due to wide confidence intervals which include the possibility of harm as well as benefit and no effect.

Summary of findings 7. Summary of findings table - 30° head of bed elevation compared to 45° head of bed elevation for pressure injury prevention in adult patients

30° head of bed elevation compared to 45° head of bed elevation for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: intensive care unit

Intervention: 30° head of bed elevation

Comparison: 45° head of bed elevation

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with 45° head of bed elevation	Risk with 30° head of bed elevation				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	0 per 1000	-- per 1000 (-- to --)	Not estimable	120 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b}	
Health-related quality of life - not measured				-	-	
Procedural pain - not measured				-	-	
Patient satisfaction - not measured				-	-	
Costs - not measured				-	-	
Incremental costs - not measured				-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_460555336761538405.

^a One included study: Ghezeljeh, 2017: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 4: measuring the outcome; and Domain 5: selection of reported results). Downgraded one level for study limitations (risk of bias) as one study had 'high' ROB.

^b Downgraded one level for imprecision because no events occurred in either group. Downgraded one level due to small sample size.

Summary of findings 8. Summary of findings table - Prone position with lung recruitment manoeuvres compared to supine position for pressure injury prevention in adult patients

Prone position with lung recruitment manoeuvres compared to supine position for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: intensive care unit

Intervention: prone position with lung recruitment manoeuvres

Comparison: supine position

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with supine position	Risk with prone position with lung recruitment manoeuvres				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	133 per 1000	607 per 1000 (308 to 1000)	RR 4.55 (2.31 to 8.98)	116 (1 RCT)	⊕⊕⊕⊖ Low ^{a,b}	
Health-related quality of life - not measured				-	-	
Procedural pain - not measured				-	-	
Patient satisfaction - not measured				-	-	
Costs - not measured				-	-	
Incremental cost - not measured				-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_460555348028225387.

^a One included study: Zhou, 2014: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 1: randomisation; Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results).

^b Downgraded one level for study limitations (risk of bias) as the study had 'high' ROB. Downgraded one level for imprecision due to small sample size.

Summary of findings 9. Summary of findings table - Micromovement compared to standard care for pressure injury prevention in adult patients

Micromovement compared to standard care for pressure injury prevention in adult patients

Patient or population: pressure injury prevention in adult patients

Setting: operating room

Intervention: micromovement

Comparison: standard care

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with standard care	Risk with micromovement				
Pressure injury occurrence: category/stage 1 to 4 (Pressure injury category/stage 1 to 4)	92 per 1000	26 per 1000 (10 to 61)	RR 0.28 (0.11 to 0.67)	477 (2 RCTs)	⊕⊕⊕⊕ Low ^{a,b}	
Health-related quality of life - not measured				-	-	
Procedural pain - not measured				-	-	
Patient satisfaction - not measured				-	-	
Costs - not measured				-	-	
Incremental cost - not measured				-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_459938148890743647.

^a Two included studies: 1) Jing 2023: overall ROB assessed as having 'Some concerns' in one domain for this result (Domain 5: selection of reported results); 2) Zhang 2024: overall ROB assessed as 'High' with concerns in multiple domains that substantially lowers the confidence in the results (Domain 1: randomisation; Domain 2: deviations from intended intervention; Domain 3: missing outcome data; ways in one domain for this result; Domain 4: measuring the outcome; and Domain 5: selection of reported results). Downgraded one level for study limitations (risk of bias) as one study had 'high' ROB and for one study, we had 'some concerns'.

^b Downgraded one level for imprecision due to small sample size.

BACKGROUND

Description of the condition

According to the National Pressure Injury Advisory Panel (NPIAP), the European Pressure Ulcer Advisory Panel (EPUAP), and the Pan Pacific Pressure Injury Alliance (PPPIA) [1, 2, 3, 4], a pressure injury (also known as pressure sore, pressure ulcer, or bedsore) is "localised damage to the skin and/or underlying tissue, usually over a bony prominence or related to a medical or other devices, resulting from prolonged pressure or pressure in combination with shear" [5]. Prolonged mechanical loading causes deformation of the skin, blood, and lymphatic vessels, and underlying tissue, which can result in a pressure injury, especially when skin contacts support surfaces or devices [4, 6]. Immobility, malnutrition, and sensory loss can increase pressure injury risk in all patients, especially individuals who are older, critically ill, or those with neurological conditions and injuries. Individual tissue tolerance to prolonged mechanical loading varies according to factors such as tissue type, age, disease, health status, and skin temperature [6]. In keeping with the 2019 [3] and 2025 International Clinical Practice Guidelines [4], throughout this review, we will use the term pressure 'injury' rather than 'ulcer', although the official term used by the WHO (World Health Organization) in the ICD-11 (International Classification of Diseases) for Mortality and Morbidity Statistics is "EH90 Pressure ulceration" [7].

Pressure injuries range from intact red skin through to deep tissue ulceration with exposed muscle, bone and tendon [3] (Supplementary material 8). Using the classification system outlined in the international guidelines, pressure injuries are classified according to the amount of visible tissue loss and palpatory identification of underlying structures [3]. The words 'stage' [8], 'grade', and 'category' are used interchangeably [3] to describe the levels of tissue loss. In keeping with the most recent guidance and for consistency [3], we have used the most frequently used term 'category/stage' throughout this review. Pressure injuries are classified from category/stage 1 to 4, along with two additional classifications: 'unstageable' and 'suspected deep tissue injury' [3]. A category/stage 1 pressure injury has intact skin with localised and persistent non-blanching erythema (redness), which may not be visually obvious in individuals with darker skin tones [3]. A category/stage 2 pressure injury involves partial-thickness skin loss (epidermis and dermis) [3]. A category/stage 3 pressure injury involves full-thickness skin loss involving damage, or necrosis, of subcutaneous tissue, whereas in a category/stage 4 pressure injury, the damage extends to the underlying bone, tendon, or joint capsule [3]. Classifying pressure injuries is subjective in nature, yet evidence indicates clinician classification agreement ranges from good to excellent [3].

Pressure injuries can impact an individual's quality of life due to pain, discomfort, and wound odour, causing psychological impacts, social isolation, and, for some, a pressure injury will result in death [9]. For healthcare systems, pressure injuries lead to higher economic costs due to longer episodes of care, complex pressure injury treatment, and increased clinician workload [10].

Pressure injuries are a patient safety issue that occurs in many healthcare settings (e.g. acute, aged care (also known as nursing home), and rehabilitation facilities, intensive care units, emergency departments), and they are considered preventable adverse events [3, 11]. The healthcare setting is also used to describe a pressure

injury. For example, when an individual living in an aged care facility develops a pressure injury, this will often be called a nursing home acquired pressure injury (NHAPI). Aged care facilities, such as nursing homes, provide care for older people, often 65 years of age or over. A 2023 systematic review and meta-analysis of six studies conducted in nursing homes found the pooled estimate of NHAPI was 8.5% (95% CI 4.4 to 13.5%) [12]. Hospital-acquired pressure injuries (HAPIs), a type of hospital-acquired complication, are amongst the top five adverse events reported globally and their prevention and treatment are widely researched [13, 14]. Estimates of HAPI incidence range from less than 3% to over 30% in medical, surgical, palliative, emergency, and critical care hospitalised patients [15, 16]. A 2020 systematic review and meta-analysis of 42 studies found the pooled pressure injury prevalence amongst hospital adult patients was 12.8% (95% CI 11.8 to 13.9%), and the pooled HAPI incidence was 8.4% (95% CI 7.6 to 9.3%) [17]. The similar incidence rates of pressure injuries in nursing homes and hospitals confirms the widespread burden of these adverse events and the need for effective prevention strategies [12, 17].

There is a paucity of evidence on the cost of treating HAPIs in any healthcare setting, which varies globally yet represents a considerable financial burden [3]. A 2020 cost-of-illness study undertaken in Australia estimated the cost of HAPIs at AUD \$9.11 billion (95% CI 9.0 to 9.2), with increased length of hospital stay and treatment accounting for most of these costs [18]. In 2016, the cost of HAPIs in the USA was estimated to exceed USD \$26.8 billion [10], while in Europe, the cost of treating a single pressure injury is up to 70,000 euros [3]. A 2025 umbrella review of 14 systematic reviews on economic evaluations of prevention and treatment interventions for pressure injuries in a range of healthcare settings, concluded that there was insufficient evidence on the cost-effectiveness of repositioning, and more research is needed [19].

Immobility, extended periods of lying and sitting on a body part, malnutrition, older age and loss of sensation are major risk factors for pressure injury development in patients [3, 20, 21]; however, older individuals and those with severely compromised states of health are particularly at risk [21]. Screening tools based on individuals' levels of activity, sensation, and mobility scores have been widely used for the assessment of pressure injury risk [22, 23]. Various prevention interventions are used and are believed to reduce the incidence of pressure injuries, with varying levels of supporting evidence, including different mattresses and overlays, prophylactic dressings, and regular repositioning [24, 25].

Description of the intervention and how it might work

Regular repositioning (i.e. turning people to change their body position to offload or redistribute pressure) has long been a fundamental and recommended component of pressure injury prevention programmes [3, 4]. Repositioning regimens, undertaken at regular intervals, are used to redistribute pressure between the body tissue and the support surface [6, 26]. The 90° lateral position has been shown in laboratory studies to decrease blood flow and transcutaneous oxygen tension close to anoxic levels (extremely low levels of oxygen) and to increase interface pressure [27]. Conversely, this does not appear to be the case when the patient is placed in a 30° lateral inclined tilt position [28]. Repositioning is regarded as also important for the prevention of other complications associated with prolonged immobility, such as pneumonia, joint contractures, and urinary tract infections [25].

Micromovement is the small incremental shifts in body weight and position, often used with repositioning devices, that aim to temporarily offload pressure [4]. This short-term strategy is used to supplement, not replace, regular repositioning regimens [4]. Micromovement is appropriate for critically unstable patients who cannot tolerate repositioning or when patients are placed in a position (e.g. prone) for surgical procedures [4]. Repositioning regimens include the schedule (or frequency, e.g. 2-hourly) and body positioning using tilt or head of bed elevation (e.g. 30°, 45°, and 90°) and/or position (i.e. regular position changes, e.g. lateral, supine, prone, micromovement).

Almost a century ago, 2-hourly repositioning was recommended as part of patient care [29]. Contemporary best-practice guidelines developed in Europe, the USA, and Australia advocated for routine repositioning of people at risk of pressure injury, with earlier versions of the guidelines also advocating 2-hourly repositioning [3, 5, 8, 30]. These recommendations were based on small observational studies (not randomised controlled trials (RCTs)) conducted up to 60 years ago, which either compared different repositioning schedules or repositioning schedules with no manual repositioning [27, 31, 32, 33]. While these studies remain useful, clinical innovation (e.g. pressure redistribution support surfaces) and advances in pressure injury research over the past two decades mean today's clinical decision-making and the selection of pressure injury prevention strategies is complex.

Sustained pressure, from lying or sitting on a particular part of the body, results in oxygen deprivation to the area [3, 6]. This can cause localised skin and tissue pain and discomfort, which stimulates the person to change position. However, if the person is physically, cognitively, or medically unable to reposition themselves, or has impaired sensation and therefore does not experience the discomfort, repositioning assistance will be required. Repositioning removes or reduces the duration of pressure on the tissues and so decreases tissue hypoxia [3, 6], and consequently theoretically decreases the risk of pressure injuries [26].

Negative aspects of frequent repositioning

Whilst regular repositioning regimens underpin current clinical practice, it may also be associated with negative consequences for patients, nursing staff, and healthcare organisations [29, 34, 35, 36, 37, 38, 39, 40]. For patients, repositioning can lead to sleep disruption, in particular, sleep fragmentation [38]. In acutely ill people, disruption of sleep can lengthen recovery, suppress immune function, and predispose them to infection [36, 39]. A sleep cycle, which has light and deep stages of sleep, occurs about every 90 minutes. Consequently, if repositioning is undertaken every two hours, it may result in fragmentation of sleep at a detrimental stage of the sleep cycle [29, 37]. Other negative effects of repositioning include possible increases in patients' pain perception. Although regular movement is important, unnecessary repositioning may cause increased discomfort for people with wounds, stiff joints, bone pain, or contractures [29, 38].

For nurses, frequent patient repositioning can result in musculoskeletal disorders at a rate exceeding that of workers in construction, mining, and manufacturing [35, 38, 38]. These injuries are attributed partly to repeated manual patient-handling activities, often associated with repositioning patients and working in extremely awkward positions [35, 40]. Back pain and injury have

a major impact on the efficiency of the nursing workforce [41]. In 2020, musculoskeletal back sprains and strains amongst private industry registered nurses in the United States ranked second in the reported non-fatal occupational injury and illness involving days away from work [42]. Back injuries and the resultant workers' compensation claims for nurses are expensive [37]. For example, in Australia, for the period 2008 to 2023, the highest number of workers' compensation claims were lodged by the healthcare and social assistance industry, with the mining and construction industries ranking second and third, respectively [43]. In total, AUD \$2.5 billion (2008 to 2023) has been paid to healthcare workers in claims for these injuries [43]. Reducing the amount of manual handling undertaken by nurses when repositioning patients could have major benefits for nurses and hospitals.

Why it is important to do this review

For patients, pressure injuries can be painful, distressing, and life-threatening (leading to infection, sepsis, and even death), yet in many cases they are preventable [3, 11]. The substantial economic burden of pressure injuries for healthcare systems means any reduction in pressure injury rates and subsequent cost savings could be redirected to other priority areas. Repositioning regimens are often used in pressure injury prevention programmes to offload or redistribute pressure between the body tissue and a support surface [4, 6, 26]. These strategies have major implications for individuals receiving care in all healthcare settings and warrant investigation.

Whilst the potential negative aspects of repositioning have been described, the magnitude of any benefits are also uncertain, as is the optimum frequency of repositioning and the best position. Notably, the 2025 pressure injury prevention clinical practice guidelines suggest either 2- or 3-hourly repositioning intervals, with this recommendation based on empirical evidence of very low certainty [4]. We considered an update of our review to be essential due to ongoing uncertainty around optimal repositioning regimens and the emergence of new studies since the last review. Incorporating recent findings may help strengthen clinical decision-making if the cumulative research can provide clearer guidance on effective, patient-centred prevention strategies.

OBJECTIVES

To evaluate the benefits and harms, and cost-effectiveness, of repositioning regimens (i.e. repositioning frequencies, positioning, micromovement) for pressure injury prevention in adults in acute, long-term, or aged healthcare settings, compared to standard care or another regimen.

METHODS

Changes in the 2026 update

- We excluded studies where medical-device-related pressure injuries were the primary or secondary outcomes.
- We included papers on micromovement, as a positioning regimen, to align with the repositioning descriptors included in the International Clinical Practice Guideline produced in 2025 by the National Pressure Injury Advisory Panel, European Pressure Ulcer Advisory Panel and Pan Pacific Pressure Injury Alliance [4].

- We added category/stage to describe a pressure injury in keeping with the various classification systems available worldwide.
- For the 11 included studies, and for the outcomes in the summary of findings tables, we assessed risk of bias (RoB) using the revised Cochrane RoB 2 tool [44] instead of the Cochrane RoB 1 tool [45] that was used in the previous versions of this review.
- In keeping with consumer engagement and involvement in this review, we examined patient and public involvement in the included studies using the Guidance for Reporting Involvement of Patients and Public (GRIPP2) checklist [46].
- We modified the wording of our objective.

For changes to protocol in previous versions, please see [Table 1](#).

Criteria for considering studies for this review

Types of studies

Any RCT that used a method of random allocation of adult patients (without an existing pressure injury at baseline) between two or more alternative repositioning regimens for pressure injury prevention was eligible. We also included cluster-randomised trials (c-RCTs), irrespective of the cluster group (i.e. patient, nurse, hospital). We excluded cross-over trials (even if randomised) and quasi-randomised studies (e.g. studies where treatment allocation was alternate or by date of birth).

The review of health economic evidence included comparative full and partial economic evaluations conducted within the framework of eligible RCTs (i.e. cost-effectiveness analyses, cost-utility analyses, cost-benefit analyses, and cost-analyses of a repositioning intervention and a relevant comparator), as well as RCTs and c-RCTs reporting more limited information, such as estimates of resource use or costs associated with repositioning and a comparator. The review considered only health economics studies conducted alongside effectiveness RCTs.

Types of participants

Any adult, without an existing pressure injury, admitted to any acute, long-term, or aged healthcare setting.

Types of interventions

We anticipated that comparisons would include repositioning regimens compared with other standard practices or with alternative repositioning regimens, regardless of the inclusion of standard clinical care strategies (e.g. type of support surfaces used). We included studies that evaluated one or more of the following comparisons.

- Comparisons between different repositioning frequencies, e.g. 2-, 3-, 4-, 6-hourly repositioning, etc, where the only systematic difference between groups was the frequency of repositioning
- Comparisons between different positions for repositioning, e.g. 30° versus 45° head of bed elevation, 30° recumbent tilt versus 90° lateral rotation, where the only systematic difference between groups was the positioning
- Comparisons of micromovement with standard care
- Comparisons of the repositioning regimen with standard practice (as defined by the trial author/s)

Outcome measures

We collected data on the listed outcomes for the duration of each included study.

Critical outcomes

The cumulative incidence of pressure injury (i.e. proportion of participants with a new pressure injury) of any category/stage using previously defined criteria [1, 3, 5, 8, 30] or as defined by the trial authors, anywhere on the body, following recruitment into the study. Studies where the pressure injury incidence was not included as a primary or secondary outcome were excluded. Medical-device related pressure injuries were also excluded.

Important outcomes

- Health-related quality of life (HRQoL), including utility scores (as reported by the trial author/s)
- Procedural pain (as reported by the trial author/s)
- Patient satisfaction (as reported by the trial author/s)
- Costs: including cost of pressure injury prevention; cost of related health practitioner time or visits; cost avoided by pressure injury prevention (e.g. treatment cost per patient per pressure injury wound; cost to treat adverse events, infections, or complications of pressure injuries; duration or cost of hospital stay for pressure injury wound healing, adverse events, and complications; indirect costs to society associated with pressure injury, such as lost productivity)
- Incremental cost per event avoided, such as per additional pressure injury prevented; incremental cost per life year gained; incremental cost per quality-adjusted life year (QALY) gained, and cost-benefit ratio

Search methods for identification of studies

We replicated the search strategy used in the 2020 review [47]. Prior to running the search for this 2026 update, the strategy was reviewed by the Information Specialists from the Cochrane Central Executive Team.

Electronic searches

On 7 May 2025, we searched the electronic databases listed below to identify reports of relevant clinical trials. The search results were downloaded into Endnote [48] and then uploaded to the Covidence systematic review web platform [49].

- Cochrane Wounds Specialised Register (database archived, so no searches undertaken)
- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 5) in the Cochrane Library
- Ovid MEDLINE (RCT)(searched 13 February 2019 to 7 May 2025)
- Ovid MEDLINE (health economics)(searched 13 February 2019 to 7 May 2025)
- Embase (RCT via Elsevier) (searched 13 February 2019 to 7 May 2025)
- Embase (health economics) (searched 13 February 2019 to 7 May 2025)
- EBSCO CINAHL Complete (Cumulative Index to Nursing and Allied Health Literature; searched 13 February 2019 to 7 May 2025)

- NHS Economic Evaluation Database (NHS EED) via York database (searched 13 February 2019 to 7 May 2025)

The search strategies for the electronic databases are shown in [Supplementary material 1](#). We combined the Ovid MEDLINE (RCT) search with the Cochrane Highly Sensitive Search Strategy for identifying randomised trials in MEDLINE: sensitivity- and precision-maximising version (2008 revision) [50]. We combined the Embase search with the Ovid Embase filter developed by the UK Cochrane Centre [50]. We combined the CINAHL Complete searches with the trial filters developed by the Scottish Intercollegiate Guidelines Network [51]. There were no restrictions with respect to language, date of publication, or study setting. We combined Ovid MEDLINE (RCT), Embase, and EBSCO CINAHL Complete searches with filters developed by the Centre for Reviews and Dissemination for the identification of economic studies [52].

We also searched the following clinical trials registries.

- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (www.clinicaltrials.gov) (searched 7 May 2025)
- World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) (apps.who.int/trialsearch/Default.aspx) (searched 7 May 2025)
- EU Clinical Trials Register (www.clinicaltrialsregister.eu/ctr-search/search) (searched 7 May 2025)

Search strategies for clinical trial registries are shown in [Supplementary material 1](#).

We did not restrict searches by language, study setting, or publication status. We made every effort to obtain translations of papers that were not published in English.

Searching other resources

Details of the search strategies used for the previous versions of this review [47, 53]. We searched for any post-publication amendments for included or eligible studies. During this review, we checked for retractions, expressions of concern, errata, and corrigenda by searching the Retraction Watch database, reviewing journal notices in which included studies were published, and screening PubMed's 'Errata', 'Corrigendum', and 'Retraction' tags. These checks ensure problematic studies are identified and addressed.

Searching reference lists of included trials and relevant reviews

We manually searched the reference lists of the included trials, as well as relevant systematic reviews, meta-analyses, and health technology assessment reports, in order to identify any other potentially eligible trials or ancillary publications.

Searching by contacting individuals or organisations

We contacted authors of key papers and abstracts to request further information about their trials when necessary.

Conference proceedings

We searched for conference abstracts from the following professional organisations: the American Professional Wound Care Association, Wounds Australia, and the European Wound Management Association. Our searches were carried out in Ovid

MEDLINE, Embase, and EBSCO CINAHL Complete for the period 13 February 2019 to 7 May 2025.

Adverse effects

We did not perform a separate search for adverse effects of interventions used; we considered adverse effects described in the included studies only.

Data collection and analysis

We performed the searches, screening, data collection, and data analysis according to the methods stated in the published protocol for this review [54], which were based on the *Cochrane Handbook for Systematic Reviews of Interventions* [55]. We used the Guidance for Reporting Involvement of Patients and Public (GRIPP2) checklist [46] to examine consumer involvement in the included clinical studies.

Selection of studies

After uploading the studies to the Covidence systematic review web platform [49], the automation tool marked ineligible studies. Following training on the review inclusion and exclusion criteria, two review authors (SLL, WC) independently assessed all titles and abstracts. We retrieved the full-text reports of all potentially relevant trials for further assessment of eligibility based on the inclusion criteria. Differences of opinion were resolved by consensus or by referral to a third review author (BG). We were not blind to study authorship. We recorded reasons for exclusion.

Data extraction and management

Four review authors (SLL, BG, SP, SL) independently extracted data from eligible studies using a pre-designed data collection tool. Prior to data extraction, all review authors underwent training. The data extraction tool was pilot tested by SLL and BG using one of the included articles. Any differences in data extraction were first discussed between the review authors with the aim of reaching consensus. A fifth review author (EM) adjudicated differences of opinion. For studies that included an economic component, MD (Health Economist) and SLL extracted the relevant data. We included studies published in duplicate, but extracted data to ensure that information was not missed, and we identified the primary reference for the purpose of this review. If data were missing from reports, we attempted to contact the trial authors via email to obtain the missing information or to clarify intervention descriptions. One review author (SLL) entered the data into Review Manager (RevMan) software [56], and two review authors (BG, LT) checked the data for accuracy. We extracted the following information.

- Author, title, journal title, year of publication, country
- Study aim
- Study design
- Healthcare setting
- Study start and end dates
- Inclusion/exclusion criteria
- Sample size
- Participant consent
- Participant characteristics by treatment group including number of eligible and randomised people

- Completeness of follow-up and treatment of missing values (e.g. use of intention-to-treat, per-protocol, or other imputation method)
- Intervention details
- Types of outcome measures in relation to primary (percentage of new pressure injury) and secondary outcomes
- Duration of follow-up
- Patient and public involvement
- Analysis
- Results and conclusions relevant to review
- Conflict of interest
- Ethics approval
- Funding sources

For economic studies, we extracted additional data in relation to the following.

- Estimates of specific items of resource use per person
- Estimates of unit costs (extracted separately to resource use)
- Price year and currency
- Decision-making jurisdiction
- Analytic perspective
- A point estimate and a measure of uncertainty (e.g. standard error or confidence interval) for measures of incremental resource use, costs, and cost-effectiveness, if reported
- Details of any sensitivity analyses undertaken, and any information regarding the impact of varying assumptions on the magnitude and direction of results

Risk of bias assessment in included studies

We used the 2008 version of Cochrane's risk of bias (RoB 1) tool [45] to assess the RoB in the 2014 [53] and 2020 [47] reviews, with the previous criteria outlined in those documents. Following recent recommendations in the *Cochrane Handbook for Systematic Reviews of Interventions* [55], we used the revised Cochrane RoB 2 tool to assess RoB in this 2026 update [44]. As there were only eight studies included in the 2020 version, and we did not expect many new studies, the review authors and Cochrane group decided to assess all included studies using the RoB 2 tool [44] because this would improve consistency and quality. Guided by the Revised Cochrane RoB 2 tool for randomised trials short version (cribsheet) [44], review authors (SLL, BG, SP, SL) independently assessed the three trials in the 2026 review and eight trials from the previous Cochrane review [47]. Any disagreements in assessment were discussed between the review authors, with a third author (EM) adjudicating differences of opinion. Where clarification was needed, we contacted the trial authors and asked open-ended questions about the study.

The RoB 2 tool addresses preliminary potential bias considerations of each study included in a meta-analysis through the use of signalling questions across five domains plus an overall RoB based on pre-determined criteria [44] ([Supplementary material 9](#)).

- Domain 1 - randomisation process
- Domain 2 - deviations from intended interventions (ITT) OR deviations from adhering to an intervention (PP (per protocol))
- Domain 3 - missing outcome data
- Domain 4 - measurement of the outcome

- Domain 5 - selection of the reported result
- Overall risk of bias (low, some concerns, high)

Responses to the domain signalling questions were either yes, probably yes, no information, probably no, no, or not applicable. We used verbatim statements from the trials to justify each response. Using the algorithm decision tree, a RoB category was assigned to each domain (i.e. low risk, some concerns, or high risk) [44], with overall RoB judged as follows.

- "Low risk of bias: The study is judged to be at low risk of bias for all domains for this result.
- Some concerns: The study is judged to raise some concerns in at least one domain for this result, but not to be at high risk of bias for any domain.
- High risk of bias: The study is judged to be at high risk of bias in at least one domain for this result **OR** The study is judged to have some concerns for multiple domains in a way that substantially lowers confidence in the result." [44].

Prior to assessment, we prespecified the effect of interest. We reported the effect of assignment to the intervention, obtained through an intention-to-treat (ITT) analysis. Where this was not reported, we assessed this as a deviation from adhering to the intended intervention under domain 2 of the RoB 2 tool (deviations from intended interventions), signalling question 2.6, and updated our RoB judgements accordingly. Using the reported data, we assessed the primary (proportion of participants with a new pressure injury) and secondary outcomes (HRQoL, procedural pain, patient satisfaction). In assessing the RoB, the review authors were not blinded to the names of trial authors, institutions, or journals.

For the economic evaluations we included, we did not conduct an assessment of the methodological quality or evidence certainty and only provide a commentary on the findings [57].

We presented our detailed RoB assessment for each study either in the text description or in the analyses (Analysis 1.1; Analysis 2.1; Analysis 3.1; Analysis 4.1; Analysis 6.1; Analysis 7.1; Analysis 9.1; Analysis 5.1; Analysis 8.1). When considering the treatment effects, we considered the overall RoB judgements as part of the GRADE methodology to assess the certainty of the evidence [58].

Measures of treatment effect

We reported effect estimates for dichotomous outcomes (e.g. the proportions of people developing new pressure injuries during follow-up) as risk ratios (RR) with 95% confidence intervals (CI). RR is calculated as the proportion of participants developing pressure injuries in the experimental group divided by the proportion in the control group. It indicates the relative likelihood of new pressure injury development under the experimental intervention (e.g. turning frequency, positioning, or micromovement) compared to standard care. An RR of 1 indicates no difference between groups in pressure injury incidence. An RR of less than 1 suggests a benefit of the experimental intervention, while an RR greater than 1 favours the control group [59]. We used the RR rather than odds ratio (OR), since ORs may be misinterpreted as RR and can give an inflated impression of the effect size when event rates are greater than 20% [59].

We have presented a tabulated analysis of the identified economic data in accordance with advice outlined in the CHEERS

(Consolidated Health Economic Evaluation Reporting Standards) checklist [60]. We tabulated the main characteristics and results of the identified economic evaluation studies and expanded on these narratively.

Given the likely lack of direct comparability in resource use and cost data between different healthcare contexts and settings, we did not pool economic outcomes for any of the included studies, but instead incorporated a discussion of key drivers and the impact of assumptions on the available economic evaluations, scenarios that are likely to lead to the most and least cost-effective use of repositioning for pressure injury prevention, as well as guidance on future research that might be required to assess the economic value of repositioning as an intervention for pressure injury prevention.

Costs

For economic studies, all substantial costs that were observed to differ between people repositioned for pressure injury prevention and people administered the comparator treatment were captured and reported as part of the review of economic evaluations [57].

We reported resource utilisation and unit costs separately, along with the currency and price year in each original study. These costs were converted to 2018 values by employing a web-based conversion tool that applies implicit price deflators for the gross domestic product (GDP) of that currency and then converts into the currency most frequently observed in the articles reviewed using purchasing power parities (PPP) [61].

The main costs were those associated with the development of pressure injuries, specialist and other practitioner costs as measured by time or number of visits, potential cost-savings from a change in the number of hospital-bed days, and costs stemming from differing rates of adverse events and complications (including procedures initiated due to the failure of wounds to heal, such as amputation). We identified key cost drivers that would enable users of the review to gain a clear understanding of the nature of resource use associated with repositioning for pressure injury prevention.

Health state utility weights

We evaluated changes in HRQoL attributable to the intervention across included trials, using utility-based measures suitable for economic evaluation. We extracted data on HRQoL using multi-attribute utility instruments (MAUI) such as EQ-5D (EuroQol Five-Dimension), SF-6D (Short-Form Six-Dimension), or HUI (Health Utilities Index), as well as direct elicitation methods such as the time trade-off, standard gamble time trade-off and standard gamble [62]. This enabled assessment of the utility data for comparability and representativeness, considering issues such as the stages of pressure injury, patient population, timing of the baseline point and follow-up collection, MAUI used, and algorithm for scoring the MAUI. We evaluated heterogeneity in measurement approaches and potential biases arising from instrument selection or valuation methods. We included a discussion on the direction and magnitude of utility changes attributable to the intervention, their clinical relevance, and implications for economic evaluations, particularly in terms of quality-adjusted life year (QALY) gains or losses [63].

Unit of analysis issues

For the included trials, we extracted and analysed the single new pressure injury outcome per person recruited in the study, taking into account the level at which randomisation occurred. For studies with a parallel-group design, we collected and analysed a single measurement for each outcome for each person. In instances where there were multiple observations per person for the same outcome, we contacted the trial author(s) to clarify how these data were handled.

For c-RCTs where clustering was not taken into account in the study analysis, we considered adjusted sample sizes using the methods described in Chapter 23 of the *Cochrane Handbook for Systematic Reviews of Interventions* [64]. However, the best estimate of a relevant intraclass correlation coefficient (ICC) for estimating the design effect was so small (such as 0.001) that we used the original reported study data without adjustment.

Dealing with missing data

In the first instance, we contacted study author(s) to obtain missing outcome data. We performed an available-case analysis that only included participants with reported outcome data. We considered this approach to be more conservative than assuming that participants lost to follow-up did not develop a pressure injury. Where standard deviations (SD) were missing, we imputed them from other studies or computed them from standard errors (SE) using the formula $SD = SE \times \sqrt{N}$, where these values were available [55]. We did not conduct best-case and worst-case analyses [55].

Reporting bias assessment

There were too few studies to generate funnel plots, but had there been at least 10 studies in a meta-analysis, we would have generated funnel plots to assess potential publication bias, evaluating asymmetry visually [65].

Synthesis methods

In line with the *Cochrane Handbook for Systematic Reviews of Interventions* [55], we assessed and reported heterogeneity using a combination of clinical, methodological, and statistical considerations. We assessed clinical heterogeneity by examining differences across studies in participant characteristics, intervention types, and intervention durations, and measured outcomes. We assessed methodological heterogeneity by comparing study designs and the risk of bias across included trials. For cluster trials, we evaluated outcomes at the same level as group allocation [55]. When we judged studies to be clinically and methodologically similar, and outcome measures were comparable, we considered them suitable for pooling in a meta-analysis.

We quantified statistical heterogeneity using the Chi² test (with a significance level set at $P < 0.10$) and the I² statistic, which describes the percentage of variability in effect estimates that is due to heterogeneity rather than chance. We considered an I² value above 75% to indicate substantial heterogeneity. However, we did not rely solely on statistical tests to determine heterogeneity, as these may fail to detect important differences. Instead, we combined statistical indicators with clinical and methodological assessments to guide our decisions [66].

Where appropriate, we pooled outcome data using either the fixed-effect or random-effects model in RevMan [56], and presented the pooled effect estimates with 95% CI.

- We used the fixed-effect model when studies were judged to have minimal clinical and methodological heterogeneity, and the assumption of a common underlying treatment effect was reasonable. Fixed-effect meta-analysis provides a more precise estimate but assumes that any differences between studies are due to chance alone.
- We used the random-effects model when there was evidence of between-study heterogeneity that could not be readily explained. This approach assumes that the true effect sizes vary across studies and incorporates this variation into the confidence intervals, which tend to be wider [66].

We used forest plots to visually summarise meta-analytic results [66]. For dichotomous outcomes, we reported pooled RRs with 95% CIs. Had we had continuous outcomes, we would have reported mean differences (MDs) or standardised mean differences (SMDs) with 95% CIs.

We did not plan to pool data for economic outcomes, as differences in economic evaluation methods and generalisability limit the feasibility of statistical synthesis [57].

Where statistical synthesis of data from more than one study was not possible or considered inappropriate, we conducted a narrative review of eligible studies [66]. We described the included studies in a structured narrative summary based on comparators and entered quantitative data into RevMan [56].

Using GRADEproGDT, we planned to complete summary of findings tables for comparisons with pooled data, as well as for individual studies, along with RoB 2 and GRADE assessments [65, 66, 67].

Investigation of heterogeneity and subgroup analysis

We had planned subgroup analysis to examine the effect of potentially influential factors on outcome with regard to the type of patient and care setting [66]. We did not attempt this as there was considerable heterogeneity and variation between estimates.

Equity-related assessment

We did not undertake an equity-related assessment because repositioning is considered standard clinical care for all patients and is not differentially targeted or delivered to disadvantaged populations or groups.

Sensitivity analysis

We had planned to perform sensitivity analyses where necessary to test whether the findings were robust to the method used to obtain them, and compared the results of two or more meta-analyses using different assumptions [65]. However, as we assessed six of the included trials at high RoB, this analysis was not undertaken.

Certainty of the evidence assessment

The main results are presented in the summary of findings (SoF) tables (Summary of findings 1; Summary of findings 2; Summary of findings 3; Summary of findings 4; Summary of findings 5; Summary of findings 6; Summary of findings 7; Summary of findings 8;

Summary of findings 9). The following comparisons are presented in the summary of findings tables:

- pooled data for different repositioning frequencies: 2- versus 4-hourly, 2- versus 3-hourly, 3- versus 4-hourly, and 4- versus 6-hourly (Summary of findings 1; Summary of findings 2; Summary of findings 3; Summary of findings 4);
- pooled data for different positions of the patient or hospital or bed: various degrees of tilt (30° compared to 90°) (Summary of findings 6); and
- pooled data for micromovement compared to standard care (Summary of findings 9).

In these SoF tables, we summarised the key data for the following primary outcome and secondary outcomes [58]:

- development of a new pressure injury;
- HRQoL;
- procedural pain;
- patient satisfaction;
- costs; and
- incremental cost.

The SoF tables also present our GRADE assessments of the certainty of evidence for these outcomes, which we determined using the GRADEpro GDT software [58, 68]. We assessed the certainty of evidence as high, moderate, low, or very low, reflecting our level of confidence that the estimate of effect is close to the true effect [58, 69].

- High certainty: very confident that the true effect lies close to that of the estimate of the effect.
- Moderate certainty: moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.
- Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

In order to arrive at this judgement for each comparison and outcome, one author (LT) completed the GRADE assessment, which was checked by a second author (SLL), and a third author (BG) adjudicated any disagreements. In GRADE, a body of evidence composed of RCTs begins as 'high certainty'. The level of certainty can be downgraded based on [58, 69]:

- RoB (i.e. randomisation process, deviations, missing outcome data, outcome measurement, selection of reported result and, overall RoB);
- inconsistency;
- indirectness;
- imprecision;
- publication biases [44].

We used the following decision rules for downgrading the evidence certainty:

- if we had no serious concern, we did not downgrade;

- if we had serious concern, we downgraded the evidence one level (e.g. from high to moderate (– 1)); and
- if we had very serious concern, we downgraded the evidence two levels (e.g. from high to low (– 2)) [58, 68].

As assessing the certainty of the evidence using GRADE involves consideration of the RoB criteria ([Supplementary material 9](#)) and assessment of methodological heterogeneity, we linked our Cochrane RoB ratings to the GRADEpro GDT software assessment [58, 68].

We also conducted GRADE assessments for those comparisons that did not have pooled data ([Summary of findings 5](#); [Summary of findings 7](#); [Summary of findings 8](#)), and narratively described each trial in the results section.

Consumer involvement

Guided by the Cochrane framework for engagement and involvement of patients, carers and the public [70], review author, Dinesh Palipana (DP), who is an international consumer advocate,

led the consumer involvement aspect of the review. He used his lived experience to make the review evidence relevant, sensitive, and accessible.

RESULTS

Description of studies

See [Supplementary material 2](#); [Supplementary material 3](#); [Supplementary material 4](#) and [Table 2](#). Sections of one study were published in Chinese (Zhou 2014 [71]) and translated into English by a researcher fluent in the written language.

Results of the search

In this 2026 update, we conducted a search for clinical trials of the interventions and economic evaluation substudies. The results of the two search types are reported separately. The study flow diagram of clinical studies is shown in [Figure 1](#), and the study flow diagram of economic evaluations is shown in [Figure 2](#). We assessed 707 clinical study records, which included 117 records from clinical trial registries and 245 economic evaluation records.

Figure 1. Study flow diagram for clinical studies

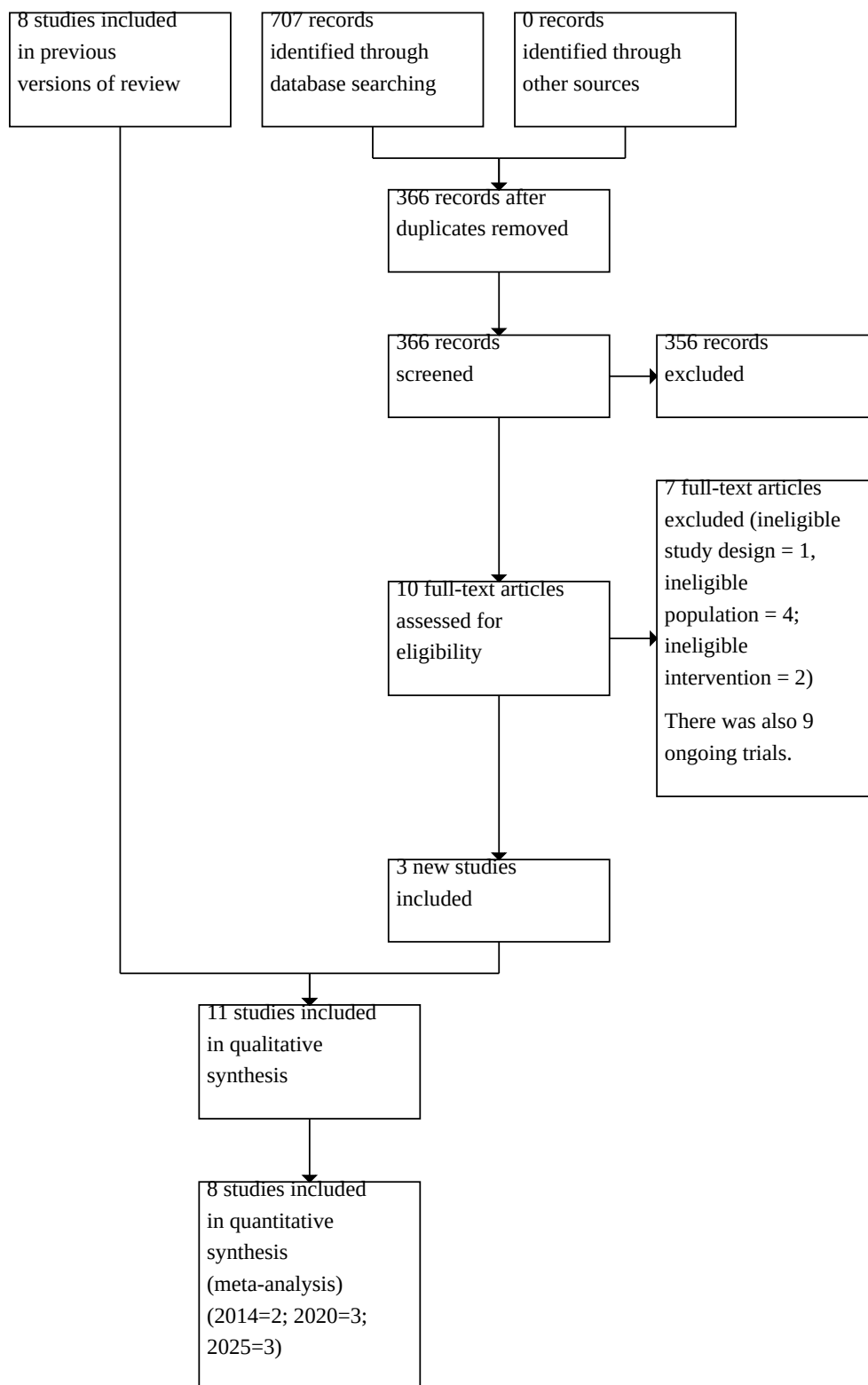
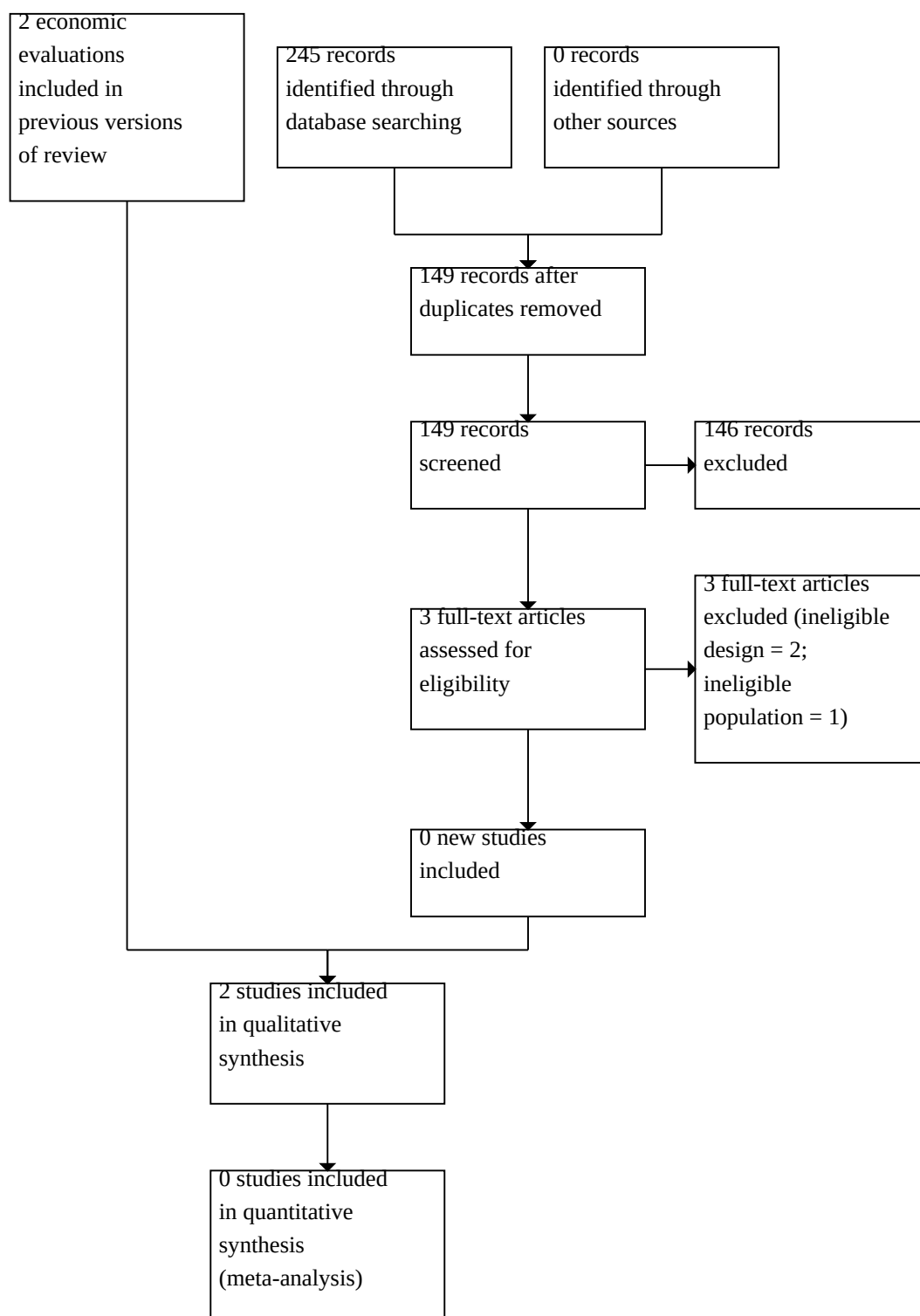


Figure 1. (Continued)

Figure 2. Study flow diagram for economic studies



For the clinical studies, Covidence automatically identified 341 duplicates [49], resulting in 366 unique records to screen. We excluded 356 records because they did not meet one or more of our inclusion criteria. We retrieved and assessed 10 full-text intervention papers, with three new clinical trials meeting the review eligibility criteria (Jin 2024 [72]; Tarigan 2021 [73]; Zhang 2024 [74]). In June 2025, we emailed the listed investigator of the Jin 2024 study to clarify the intervention description. A response was received confirming our interpretation that the intervention and control groups were eligible. Therefore, we added three new trials to the eight trials from the 2020 review [47] (Bergstrom 2013 [75]; Defloor 2005 [76]; Ghezeljeh 2017 [77]; Manzano 2014 [78]; Moore 2011 [79]; Pickham 2018 [80]; Young 2004 [81]; Zhou 2014). In total, the 2026 update includes 11 clinical trials that were conducted in adult patients across acute and aged healthcare settings, and published between 2005 and 2024. See Figure 1.

Clinical trials registries search

We identified six new ongoing clinical studies in this 2026 update (ChiCTR2100042476 [82]; ChiCTR2100046245 [83]; KCT0009366 [84]; NCT04550182 [85]; NCT04604665 [86]; NCT05549830 [87]), which we added to the three ongoing trials from the 2020 review [47] (NCT02479425 [88]; NCT03048357 [89]; NCT03454230 [90]). In May 2025, we attempted to contact the nine listed investigators to enquire about the progress of these ongoing trials. Only two investigators responded, and they advised they were finalising their analysis and writing manuscripts for submission (KCT0009366; NCT04604665). Results from these nine ongoing trials were not published by the time of publication of this review (June 2026).

Economic analysis search

Electronic searches for this 2026 update yielded 245 economic evaluation records (Figure 2). Covidence automatically identified 96 duplicates [49], leaving 149 records for us to screen; we excluded 146 records. We retrieved and assessed three full-text economic records. We excluded all three because they did not meet our inclusion criteria. Hence, there were no additional economic substudies to add to this 2026 update. We brought forward the economic substudies relating to Moore 2011 (Moore 2013 [91]) and Bergstrom 2013 (Paulden 2014 [92]) from our 2020 review [47]. See Figure 2.

Included studies

See [Supplementary material 2](#) and [Table 2](#).

We included a total of 11 clinical trials (Bergstrom 2013; Defloor 2005; Ghezeljeh 2017; Jin 2024; Manzano 2014; Moore 2011; Pickham 2018; Tarigan 2021; Young 2004; Zhang 2024; Zhou 2014) and two economic substudies (Moore 2013; Paulden 2014) in this 2026 update. Although the Moore 2011 study did not meet the review's inclusion criteria for positioning, strictly speaking (see [Types of interventions](#)), as it examined tilts in combination with repositioning, we retained this study because it addressed the review question and contributed relevant data in an area with limited available evidence.

The clinical trials were mostly single-site studies (Ghezeljeh 2017; Jin 2024; Manzano 2014; Pickham 2018; Tarigan 2021; Young 2004; Zhang 2024; Zhou 2014), other than three studies that were conducted in multiple nursing home sites (Bergstrom 2013; Defloor 2005; Moore 2011). Bergstrom 2013 recruited 27 nursing homes

across two countries (7 in Canada and 20 in the USA); Defloor 2005 conducted their study in 32 wards within 11 Belgium nursing homes, and Moore 2011 completed their trial in 12 nursing homes in Ireland.

Nine of the clinical trials were parallel-design RCTs (Bergstrom 2013; Ghezeljeh 2017; Jin 2024; Manzano 2014; Pickham 2018; Tarigan 2021; Young 2004; Zhang 2024; Zhou 2014) and two were c-RCTs (Defloor 2005; Moore 2011). Defloor 2005 undertook cluster randomisation in two 4-week periods. In the first period, a repositioning frequency (2-, 3-, 4-, 6-hourly, and standard care) was randomly allocated to each of the 32 wards within 11 nursing homes. This randomisation process was repeated for the second 4-week period. Moore 2011 randomly clustered at the nursing-home level. While Moore 2011 reported an ICC of 0.001, Defloor 2005 did not report the ICC. As such, we used the ICC from a falls study with a similar patient group and care context [93] to impute the ICC for Defloor 2005. The Defloor 2005 trial used 32 clusters based on wards, whilst the Moore 2011 trial used 12 clusters based on study site. We did not adjust sample sizes for clustering in the two c-RCTs (Defloor 2005; Moore 2011). In the Defloor 2005 trial, while a total of 521 participants were included in the trial, only 250 participants were randomised to arms relevant to this review.

Most of the clinical trials ($n = 7$) had two study arms (Jin 2024; Manzano 2014; Moore 2011; Pickham 2018; Young 2004; Zhang 2024; Zhou 2014). Three had three study arms (Ghezeljeh 2017; Bergstrom 2013; Tarigan 2021). One of the c-RCTs had five arms (Defloor 2005), but only the four repositioning arms were included in this review.

Four trials were conducted in Asia (China (Jin 2024; Zhang 2024; Zhou 2014) and Indonesia (Tarigan 2021)), two in Europe (Belgium (Defloor 2005) and Spain (Manzano 2014)), two in North America (US and Canada (Bergstrom 2013; Pickham 2018)), and one each in Iran (Ghezeljeh 2017), Ireland (Moore 2011), and Wales (Young 2004).

The studies were conducted in long-term care settings (Bergstrom 2013; Defloor 2005; Moore 2011), intensive care units (ICU) (Ghezeljeh 2017; Manzano 2014; Pickham 2018; Zhou 2014), an ICU/High Care Unit (HCU) setting (Tarigan 2021), operating theatre settings (Jin 2024; Zhang 2024), and an acute care facility (Young 2004).

Bergstrom 2013 and Moore 2011 conducted economic substudies, with the findings reported by Paulden 2014 and Moore 2013, respectively.

Types of participants

There were 4462 participants enrolled in the 11 trials (Bergstrom 2013; Defloor 2005; Ghezeljeh 2017; Manzano 2014; Moore 2011; Pickham 2018; Young 2004; Zhou 2014; Jin 2024; Tarigan 2021; Zhang 2024). The number of participants randomised in each included study ranged from 44 up to 1312. The numbers in each trial were 967 (Bergstrom 2013), 838 (Defloor 2005), 120 (Ghezeljeh 2017), 200 (Jin 2024), 329 (Manzano 2014), 213 (Moore 2011), 1312 (Pickham 2018), 44 (Tarigan 2021), 46 (Young 2004), 277 (Zhang 2024), and 116 (Zhou 2014).

Participant age ranged from 18 to 90 years. Participants in four trials, which were conducted in nursing homes, were all over 65 years of age, and assessed as having a moderate or high risk

of pressure injury development, and limited mobility (Bergstrom 2013; Defloor 2005; Moore 2011; Young 2004). Participants in the ICU trials were critically ill due to medical or surgical conditions or trauma, with or without mechanical ventilation support (Ghezeljeh 2017; Manzano 2014; Pickham 2018; Tarigan 2021; Zhou 2014). Participants in the two trials conducted in the operating theatre setting had elective (planned) posterior lumbar surgery (Zhang 2024) and abdominal surgery under general anaesthesia in the supine position (Jin 2024).

Types of interventions

Repositioning frequencies

Five of the 11 included trials assessed various combinations of repositioning frequencies (2-, 3-, 4-, and 6-hourly) (Bergstrom 2013; Defloor 2005; Manzano 2014; Pickham 2018; Tarigan 2021). The Bergstrom 2013 trial had three arms and compared 2-, 3-, and 4-hourly turning schedules in nursing home residents. Across the three intervention arms, on-time (+/- 30-minutes) repositioning was required whilst in bed. High-density foam mattresses were used as standard care for all trial participants in one trial (Bergstrom 2013).

The Defloor 2005 trial, conducted in 11 nursing homes, compared 2-, 3-, 4-, and 6-hourly repositioning using a semi-Fowler's or lateral position, in combination with a standard institutional mattress or a viscoelastic foam mattress. In this study, we excluded the large "standard care" arm comprising 576 people allocated care based on nurses' clinical judgement (a range of support surfaces but no repositioning) because it differed from the other four arms (Defloor 2005).

The Manzano 2014 trial compared 2- and 4-hourly repositioning in conjunction with 30° head of bed elevation tilt positions (alternating left/right/supine) and alternating air-pressure mattresses (standard care across patients in both groups) in ICU participants.

The Tarigan 2021 trial compared 2-, 3-, and 4- hourly repositioning in 44 patients admitted to an ICU/HDU. All participants were first placed on the left side, then supine, followed by the right side position. The left and right side positions were maintained in the Sim's position (i.e. semi-prone) with a support pillow placed at the participant's back.

The Pickham 2018 trial assessed the use of real-time wearable patient sensor data with visual reminders for 2-hourly repositioning compared to real-time patient sensor data without visual reminders and with nurse-initiated repositioning (standard care).

Position

Six of the 11 included trials compared different positions for the patient or the bed. Three trials examined tilt or head of bed elevation positions (30°, 45°, or 90°) (Ghezeljeh 2017; Moore 2011; Young 2004) in acute and nursing home settings. Two trials studied micromovement (Jin 2024; Zhang 2024) compared to standard care in the operating theatre, and one trial evaluated prone positioning compared to supine positioning in ICU patients (Zhou 2014).

The trial by Ghezeljeh 2017 compared 30° head of bed elevation tilt with a 45° head of bed elevation tilt in ICU participants at three hospitals. Participants in the intervention groups were positioned using either 30° or 45° head of bed elevation, whilst participants in

the "control" group were given the "routine" position (Ghezeljeh 2017), although this was not described in the study. All participants across the three groups were repositioned 2-hourly, underwent skin checks/care, had bed linen changes when required, and were lifted during position changes for three days following ICU admission (Ghezeljeh 2017).

In two trials (Moore 2011; Young 2004), a 30° tilt position was compared with a standard 90° supine/lateral position. Participants in both the intervention and control groups were tilted left side, back, right side, and back. The Moore 2011 and Young 2004 trials essentially compared the same tilts (30° versus 90°) and the same repositioning frequency for the 30° tilt; however, there was a difference in the frequency of repositioning overnight for the 90° tilt groups. In the Moore 2011 trial, participants in the 90° tilt group were repositioned 6-hourly overnight compared with 2- to 3-hourly overnight in the Young 2004 trial.

The Zhou 2014 trial compared a prone positioning regimen, combined with lung recruitment manoeuvres, to the standard supine position. Participants in the intervention group were positioned prone for 10 hours, then repositioned in the supine position for 4 to 6 hours, and then repositioned to the prone position (Zhou 2014). Study participants in the control group were positioned in the supine position (standard care) (Zhou 2014).

Micromovement

The Jin 2024 trial compared micromovement with standard care (silicone foam dressings). All study participants were first placed on the operating theatre table in the supine position (Jin 2024). Participants in the intervention group received a micromovement protocol (i.e. operating table tilted 15° left and right alternating every one hour until the end of surgery; tilt maintained for five minutes each time) plus standard care (silicone foam dressings applied to the sacrococcygeal area and shoulder blades only) (Jin 2024). Control group participants only received standard care (silicone foam dressings applied to the sacrococcygeal area and shoulder blades) (Jin 2024). Intraoperative temperature was kept at 36° or above for all participants (Jin 2024).

The Zhang 2024 trial compared micromovement with standard care (comprehensive positioning plus multi-layer silicone foam dressings). The comprehensive intraoperative positioning was on a one-piece prone pad with standard pressure injury prevention protocol based on the Chinese Association of Operating Room Nurses (CORN) pressure injury risk scale (Zhang 2024). The head was positioned on a U-shaped silicone gel pad to avoid facial compression; arms curved and supported, abdomen suspended, hips on memory foam, knees padded, and feet lifted off the operating table (Zhang 2024). All study participants were first placed on the operating room table in the prone position (Zhang 2024). Participants in the intervention group received a micromovement protocol (i.e. one hour after the start of surgery, the operating table was tilted 15° left for five minutes, then two hours later the table was tilted 15° to the right for five minutes, alternating every two hours until the end of surgery) plus standard care (Zhang 2024). Control group participants received standard care (comprehensive positioning plus multi-layer silicone foam dressings) (Zhang 2024).

Types of outcomes

The primary outcome in nine of the included trials was the proportion of participants developing a new pressure injury (any category/stage) (Bergstrom 2013; Defloor 2005; Jin 2024; Manzano 2014; Moore 2011; Pickham 2018; Tarigan 2021; Young 2004; Zhang 2024). These injuries were classified using the European Pressure Ulcer Advisory Panel [30], National Pressure Ulcer/Injury Advisory Panel classification systems [1, 2, 5], the AHCPR Panel for the Prediction and Prevention of Pressure Ulcers in Adults (Defloor 2005) or a study-specific Skin Condition Assessment form (Ghezeljeh 2017). In one trial (Manzano 2014), the primary outcome was the occurrence of category/stage 2 or greater pressure injury, with a follow-up period of 24 hours. Two trials reported the incidence of pressure injury and included category/stages 1 to 4 over a 28-day period (Defloor 2005; Moore 2011), whilst the Bergstrom 2013 trial reported pressure injury occurrence over 21 days. The Tarigan 2021 trial followed participants every three days over a 14-day period and reported the occurrence of a pressure injury but not the category/stage. Young 2004 used a brief follow-up period of 24 hours and reported only category/stage 1 pressure injury (i.e. non-blanchable erythema). The follow-up period in the Pickham 2018 trial was 72 hours. Two trials reported the incidence of intraoperative-acquired pressure injury (Jin 2024; Zhang 2024), which were measured and categorised/staged by operating theatre nurses at the completion of the surgery with no further follow-up.

In two trials (Ghezeljeh 2017; Zhou 2014), the occurrence of pressure injury was a secondary outcome; the primary outcome in Ghezeljeh 2017 was the incidence of ventilator-acquired pneumonia (VAP), and in Zhou 2014, clinical complications associated with adult respiratory distress syndrome (ARDS).

Excluded studies

See [Supplementary material 3](#).

In addition to the six studies excluded from the 2020 review [47] (Beuret 2002 [94]; Girard 2014 [95]; Su 2015 [96]; Taccone 2009 [97]; Vanderwee 2007 [98]; Voggenreiter 2005 [99]), we excluded an additional 10 studies in this 2026 update: seven clinical studies (De Meyer 2019 [100]; Jiang 2020 [101]; Kapp 2023 [102]; Malekara 2025 [103]; Özkan 2019 [104]; Yap 2022 [105]; Ziegler 2023 [106]) and three health economic studies (Kang 2021 [107]; Nherera 2021 [108]; Padula 2024 [109]).

In the 2014 review [53], one trial was excluded after full-text screening (Vanderwee 2007). In this trial, participants who had pre-existing stage 1 pressure injury (non-blanchable erythema) at baseline were included, whilst those who did not have non-blanchable erythema were excluded. In the 2020 [47] review, we excluded five trials after full-text review for the following reasons: the trials included participants with a pressure injury at baseline (Beuret 2002; Girard 2014; Voggenreiter 2005); the trial focused on the use of a positioning device rather than a position or repositioning (Su 2015); and the occurrence of a pressure injury was not included (i.e. measured) as a primary or secondary outcome (Taccone 2009). In this 2026 update, our exclusion of seven clinical trials was for the following reasons: three trials included participants with a pressure injury at baseline (Malekara 2025; Yap 2022; Ziegler 2023), category/stage 1 pressure injury was not a recruitment exclusion criteria (De Meyer 2019), participants were not randomised (Özkan 2019), different repositioning regimens were used with different support surfaces

without analysis controlling for the difference (Jiang 2020), and a turning and positioning system (medical device) used 2-hourly or as per clinician assessment (Kapp 2023).

In this 2026 update, we excluded three health economic studies (Padula 2024; Nherera 2021; Kang 2021). We excluded Padula 2024 because the parent trial (Yap 2022) did not meet our review inclusion criteria. Kang 2021 and Nherera 2021 were not parallel economic evaluations of a parent RCT.

Risk of bias in included studies

We have presented a summary of our RoB 2 assessment for the pooled and non-pooled studies in [Supplementary material 5](#). The following RoB 2 assessment is based on the review's outcomes rather than individual trial data.

We assessed six of the 11 trials as having an overall high RoB (Defloor 2005; Ghezeljeh 2017; Moore 2011; Tarigan 2021; Zhang 2024; Zhou 2014). The remaining five trials were assessed as having some concerns regarding their overall RoB (Bergstrom 2013; Jin 2024; Manzano 2014; Pickham 2018; Young 2004). Overall, the quality of reporting was poor due to lack of clarity and detail, as described below.

Domain 1: randomisation process

We assessed eight of the 11 studies to be at low RoB for the randomisation process (Bergstrom 2013; Defloor 2005; Ghezeljeh 2017; Jin 2024; Manzano 2014; Pickham 2018; Tarigan 2021; Young 2004), two as having some concerns (Moore 2011; Zhang 2024), and one as high RoB (Zhou 2014). Nine trials described a process for generating the random allocation list (Bergstrom 2013; Defloor 2005; Ghezeljeh 2017; Jin 2024; Manzano 2014; Moore 2011; Pickham 2018; Young 2004; Zhang 2024). Four trials used a computer-based random number generator (Defloor 2005; Jin 2024; Moore 2011; Zhang 2024). Four trials used sequentially numbered envelopes that contained a randomisation code (Bergstrom 2013; Ghezeljeh 2017; Pickham 2018; Young 2004). In Defloor 2005, randomisation also occurred over a second four-week period. During this second period, each ward used a different prevention scheme than used in the first four-week period (Defloor 2005). A randomisation table (Zhou 2014) and randomisation by drawing lots (Tarigan 2021) were used in two trials; however, the process of randomisation was not described.

Assessment of allocation concealment in the trials involved examination of whether trial authors described how the assignment sequence was protected before and until allocation. Seven of the eleven trials described the allocation concealment process (Bergstrom 2013; Ghezeljeh 2017; Manzano 2014; Moore 2011; Pickham 2018; Young 2004; Zhang 2024).

Domain 2: (a) deviations from the intended interventions or (b) deviations from adhering to the intended interventions

Six of the 11 trials (Jin 2024; Manzano 2014; Moore 2011; Young 2004; Zhang 2024; Zhou 2014) were assessed for RoB based on deviations from the intended interventions. The five remaining trials (Bergstrom 2013; Defloor 2005; Ghezeljeh 2017; Pickham 2018; Tarigan 2021) were assessed for RoB based on deviations from adhering to the intended interventions. We assessed six trials as low RoB (Ghezeljeh 2017; Jin 2024; Manzano 2014; Moore 2011; Pickham 2018; Zhou 2014), with the remaining five trials

assessed as having some concerns (Bergstrom 2013; Defloor 2005; Tarigan 2021; Young 2004; Zhang 2024). As it is hard to envisage how blinding of participants and personnel to the frequency and nature of the repositioning regimen could be possible, all trials were likely to be at risk of performance bias. In the Jin 2024 and Zhou 2014 trials, participants were unaware of group allocation, although carers were not blinded. In most trials (Bergstrom 2013; Defloor 2005; Ghezeljeh 2017; Manzano 2014; Moore 2011; Pickham 2018; Tarigan 2021; Young 2004), it was impossible to conceal the repositioning regimens or positioning from participants or carers. Ghezeljeh 2017, Jin 2024, Manzano 2014, Moore 2011, Pickham 2018 and Zhou 2014 delivered the intervention as intended. In four trials (Bergstrom 2013; Defloor 2005; Tarigan 2021; Young 2004), some participants did not receive the intervention as intended, although it was determined this did not affect the outcome. In one trial (Zhang 2024), no descriptions were provided regarding blinding or protocol delivery.

Domain 3: missing outcome data

We evaluated attrition bias by determining whether reasons for attrition or exclusion were reported, if there was re-inclusion of participants, and whether completeness of data for each main outcome was described. We assessed eight studies at low RoB (Bergstrom 2013; Ghezeljeh 2017; Jin 2024; Manzano 2014; Moore 2011; Pickham 2018; Young 2004; Zhou 2014), two as having some concerns (Defloor 2005; Zhang 2024) and one at high RoB (Tarigan 2021). In two trials (Defloor 2005; Tarigan 2021), participants were excluded from the analysis in sufficient numbers to raise the possibility of bias. Defloor 2005 excluded 77 (9.2%) of 838 randomised participants from the analysis. In the Tarigan 2021 three-arm trial, 69.0% (9/13), 57.0% (8/14), and 70.0% (12/17) of the 44 randomised participants were excluded from groups 1, 2, and 3 respectively, due to factors including death, haemodynamic instability. For the Defloor 2005 trial, we conducted a complete-case analysis (which makes no assumptions about the outcomes for participants lost to follow-up, as this was considered to be more conservative than analysing losses as if they had not sustained a pressure injury).

Domain 4: measurement of the outcome

There was considerable variability in assessment of all stages of pressure injury amongst the included trials (Bergstrom 2013; Defloor 2005; Jin 2024; Ghezeljeh 2017; Manzano 2014; Moore 2011; Pickham 2018; Tarigan 2021; Young 2004; Zhang 2024; Zhou 2014). Such variability is problematic as the use of a subjective primary outcome measure is open to ascertainment bias.

We assessed five of the 11 studies as low RoB because the outcome assessor was blinded to group allocation and the assessment used a recognised classification system for pressure injury category/stage (Bergstrom 2013; Jin 2024; Manzano 2014; Pickham 2018; Young 2004). We assessed the remaining six trials at high RoB because we judged their methods of pressure injury outcome assessment as inappropriate (Defloor 2005; Ghezeljeh 2017; Moore 2011; Tarigan 2021; Zhang 2024; Zhou 2014). These methods included non-blinding of the outcome assessor (Zhou 2014), and the use of non-blinded research staff (Ghezeljeh 2017; Tarigan 2021; Zhang 2024) or direct carer providers (nurses) (Defloor 2005; Moore 2011; Zhang 2024) to undertake the outcome assessment.

Domain 5: selection of the reported result

We assessed all studies (Bergstrom 2013; Defloor 2005; Ghezeljeh 2017; Jin 2024; Manzano 2014; Moore 2011; Pickham 2018; Tarigan 2021; Zhang 2024; Zhou 2014) as having some concerns regarding the selection of reported results.

All included studies reported all prespecified outcomes as defined in the results section of the papers. However, only one trial (Pickham 2018) had a published protocol available [110].

We planned to assess potential publication bias using funnel plots and to assess funnel plot asymmetry visually; however, as only 11 studies were included in the review and each meta-analysis included only two, three, or four studies, this was not appropriate [66]. Six of the 11 trials included declared funding sources (Bergstrom 2013; Ghezeljeh 2017; Jin 2024; Manzano 2014; Moore 2011; Pickham 2018). One of these trials reported co-funding by product manufacturers (Pickham 2018), whilst another reported that the trialist was a co-owner of a website that promotes the use of the scale that was used in the trial to assess pressure injury risk (Bergstrom 2013).

Synthesis of results

The 11 trials randomised a total of 4462 participants. None of the included studies reported consumer engagement or involvement of consumers as researchers in their trial.

We reported three main comparisons. These were various repositioning frequencies (2- versus 4-hourly, 2- versus 3-hourly, 3- versus 4-hourly, and 4- versus 6-hourly repositioning), using any support surface; different positions - a 30° tilt, 3-hourly repositioning compared with 90° tilt, 6-hourly repositioning overnight; and micromovement compared to standard care. These comparisons are presented in [Summary of findings 1](#); [Summary of findings 2](#); [Summary of findings 3](#); [Summary of findings 4](#); [Summary of findings 6](#); [Summary of findings 9](#), and [Supplementary material 6](#). We have organised findings by comparison and a priori outcome measures. Whilst there was some clinical and methodological heterogeneity, we undertook meta-analysis where there was similarity in the primary intervention following the Cochrane-recommended methodological approach [66]. Single-study analyses are presented in [Summary of findings 5](#), [Summary of findings 7](#), and [Summary of findings 8](#).

Comparison 1: repositioning frequencies

We included data from five trials with 2902 participants in this comparison (Bergstrom 2013; Defloor 2005; Manzano 2014; Pickham 2018; Tarigan 2021). Bergstrom 2013, Defloor 2005, Manzano 2014, and Tarigan 2021 all contributed to at least one pooled estimate; we analysed the other trial as a single study (Pickham 2018). We included one c-RCT in this comparison (Defloor 2005); however, we did not adjust the data for clustering as the ICC of 0.001 (from Moore 2011 and from a falls study [93]) was so small as to make no difference.

Primary outcomes

Proportion of participants with a new pressure injury (any category/ stage 1 to 4)

Frequency of repositioning: 2-hourly versus 4-hourly

We pooled four trials comparing 2-hourly with 4-hourly repositioning using the fixed-effect model ($I^2 = 27\%$) (Bergstrom 2013; Defloor 2005; Manzano 2014; Tarigan 2021). It is uncertain whether 2-hourly repositioning compared with 4-hourly repositioning used in conjunction with any support surface, increases or decreases the incidence of pressure injury (pooled risk ratio (RR) 1.05, 95% confidence interval (CI) 0.79 to 1.39; 4 studies, 1104 participants; very low certainty evidence; Analysis 1.1). We assessed the certainty of evidence using GRADE as very low; we downgraded one level for overall high risk of bias, a further level for lack of blinding of nursing staff, and one level for imprecision (wide confidence intervals) ([Summary of findings 1](#)).

Frequency of repositioning: 2-hourly versus 3-hourly

We pooled three trials comparing 2-hourly with 3-hourly repositioning (Bergstrom 2013; Defloor 2005; Tarigan 2021). Although the interventions and comparators were similar, the I^2 statistic was 62%, which suggests substantial heterogeneity. We used the random-effects model to compute the meta-analysis (pooled RR 1.10, 95% CI 0.30 to 4.08; 3 studies, 795 participants; Analysis 2.1; very low certainty evidence) [66]. It is uncertain whether 2-hourly repositioning compared with 4-hourly repositioning used in conjunction with any support surface increases or decreases the incidence of pressure injury. Our GRADE certainty of evidence assessment was very low; we downgraded one level for high RoB, one level for significant heterogeneity, and one level for imprecision (wide confidence intervals) ([Summary of findings 2](#)).

Frequency of repositioning: 3-hourly versus 4-hourly

We pooled three trials comparing 3-hourly with 4-hourly repositioning (Analysis 3.1; [Summary of findings 3](#)). Although the interventions and comparators were similar, considerable heterogeneity ($I^2 = 75\%$) was found; we used the random-effects model to compute the meta-analysis [66]. It is uncertain whether 3-hourly repositioning, compared with 4-hourly repositioning used in conjunction with any support surface, increases or decreases the incidence of pressure injury (random-effects; pooled RR 0.99, 95% CI 0.22 to 4.43; 3 studies, 776 participants; very low certainty evidence). Our GRADE certainty of evidence assessment was very low due; downgraded one level for high RoB, downgraded one level for significant heterogeneity, and downgraded one level for imprecision (wide confidence intervals) (Bergstrom 2013; Defloor 2005; Tarigan 2021).

Frequency of repositioning: 4-hourly versus 6-hourly

The Defloor 2005 c-RCT compared 4-hourly ($n = 66$) versus 6-hourly ($n = 63$) repositioning regimens with participants nursed on a viscoelastic polyurethane foam mattress. Compared with 6-hourly repositioning, 4-hourly repositioning used in conjunction with a viscoelastic polyurethane foam mattress may decrease the incidence of pressure injury (fixed-effect; RR 0.73, 95% CI 0.53 to 1.02; 1 study, 129 participants; very low certainty evidence); but the confidence interval just crosses the line of no effect and the evidence is very uncertain. Our GRADE certainty of evidence assessment was very low; we downgraded one level for high RoB,

a further level for lack of blinding of nursing staff, and one level for imprecision (wide confidence intervals) (Analysis 4.1; [Summary of findings 4](#)).

Other frequency comparisons

The Pickham 2018 trial (1312 participants, two ICUs in the USA) compared real-time wearable patient sensor data with visual reminders for 2-hourly repositioning compared to real-time patient sensor data without visual reminders and nurse-initiated repositioning (standard care) (RR 0.28, 95% CI 0.10 to 0.75; 1 study, 1312 participants; moderate-certainty evidence). The use of sensors and visual reminders probably led to fewer people with pressure injuries. Our GRADE certainty of evidence assessment was moderate; we downgraded one level for imprecision (small number of events) (Analysis 5.1; [Summary of findings 5](#)).

Secondary outcomes

Health-related quality of life

No trial reported health-related quality of life.

Procedural pain

No trial reported procedural pain.

Patient satisfaction

No trial reported patient satisfaction.

Costs

Bergstrom 2013 included within-trial cost evaluations published by Paulden 2014 following the reporting of trial outcomes.

Using 2012 costing, Paulden 2014 performed a cost-minimisation analysis comparing the costs of 3-hourly ($n = 326$) and 4-hourly ($n = 295$) repositioning to a 2-hourly ($n = 321$) repositioning schedule. The evaluation used data collected alongside the Bergstrom 2013 trial, which was undertaken with nursing home residents who were at moderate to high risk of developing a pressure injury, recruited from 20 facilities in the USA and 7 facilities in Canada. All trial participants were nursed on a high-density foam mattress. Costs were considered from the perspective of the Ontario Ministry of Health and Long-Term Care and included those associated with nursing time for repositioning and incontinent supplies. Nursing time was estimated from a small observational study undertaken alongside the trial and assumed that 20% of residents would require two nursing staff (costed as personal support workers) for each occasion of repositioning, with the remaining 80% requiring one staff member. Costs were reported in 2012 Canadian dollars. The cost of repositioning was estimated to be CAD 11.05 or CAD 16.74 lower per resident per day for the 3-hourly or 4-hourly regimens, respectively, compared to the 2-hourly regimen. The estimates of economic benefit were driven mostly by the value of freed nursing time. The analysis assumed 2-, 3-, or 4-hourly repositioning was associated with a similar incidence of pressure injury, as no difference in incidence was observed between these strategies in the Bergstrom 2013 trial. We did not undertake a GRADE assessment of this economic evaluation, so we are unable to draw any firm or general conclusions regarding the relative costs or efficiency of 3- and 4-hourly repositioning compared with 2-hourly repositioning.

Comparison 2: position

We included data from four trials with 495 participants in this comparison (Ghezeljeh 2017; Moore 2011; Young 2004; Zhou 2014). Moore 2011 and Young 2004 contributed data for the pooled analysis. We analysed two trials as single studies (Ghezeljeh 2017; Zhou 2014).

Primary outcomes

Proportion of participants with a new pressure injury (any category/ stage 1 to 4)

30° tilt 3-hourly repositioning compared with 90° tilt 6-hourly repositioning overnight

Based on the similarity in interventions, we pooled the results of two trials (Moore 2011; Young 2004) (Analysis 6.1; [Summary of findings 6](#)).

Moore 2011 and Young 2004 compared the same tilts (30° versus 90°) using similar positions for repositioning. Therefore, we pooled the data from these two studies ($I^2 = 69\%$, random-effects model). Overall, there was no clear difference in the incidence of category/ stage 1 or 2 pressure injury (persistent erythema) between 30° and 90° tilts (pooled RR 0.62, 95% CI 0.10 to 3.97; 2 studies, 259 participants; very low certainty evidence), but the evidence was very uncertain. Our GRADE certainty of evidence assessment was very low; we downgraded one level for high RoB, one level for significant heterogeneity, and one level for imprecision (wide confidence intervals).

Other tilt regimens

There were two other trials undertaken in ICU settings that included 236 participants. However, their results could not be pooled due to the differences in tilt and repositioning regimens. In addition, the occurrence of pressure injury was a secondary study outcome. The results of these studies are presented below and in single-study summary of findings tables.

Ghezeljeh 2017 compared the use of 30° 2-hourly head-of-bed elevation tilt with a 45° 2-hourly head-of-bed elevation tilt versus standard care in a three-group randomised sample of 120 ICU patients receiving mechanical ventilation. All participants were repositioned every two hours, but the study authors did not define "standard care" relative to the degree of tilt routinely used. Although these participants were assessed for pressure injury for three consecutive days following ICU admission, this was not the primary outcome of the study. The authors reported that "the incidence of pressure ulcers in the groups was documented and compared together". There were no pressure injuries reported in the trial (the authors stated that "none of the patients who were recruited in this study developed pressure ulcers...none of the patients in the groups suffered from pressure ulcers after 3 days"). Therefore, an RR could not be estimated. Our GRADE certainty of evidence assessment was very low; we downgraded one level for high RoB, one level for imprecision (no events in either group), and one level for small sample size (Analysis 7.1; [Summary of findings 7](#)).

Zhou 2014 compared the effects of prone positioning, combined with lung recruitment manoeuvres (i.e. temporary and sustained increases in airway pressure during mechanical ventilation with the aim of opening up collapsed alveoli) versus supine positioning on the development of adult respiratory distress syndrome (primary

outcome) and pressure injury (secondary outcome) in 116 ICU patients. The occurrence of pressure injury (category/stages 1 and 2) was higher in the prone group (RR 4.55, 95% CI 2.31 to 8.98; 1 study, 116 participants; low-certainty evidence) compared to the supine group. There was no clear difference in the risk of category/ stage 2 pressure injury between the two groups ($P > 0.05$). No stage 3 or 4 pressure injuries were reported in either group. Our GRADE certainty of evidence assessment was low; we downgraded one level for high RoB and one level for imprecision (small sample size) (Analysis 8.1; [Summary of findings 8](#)).

Secondary outcomes

Health-related quality of life

No trial reported health-reported quality of life.

Procedural pain

No trial reported procedural pain.

Patient satisfaction

No trial reported patient satisfaction.

Costs

Moore 2011 included within-trial cost evaluations that were published following the reporting of trial outcomes (Moore 2013).

Using 2009 costing, Moore 2013 performed a cost-effectiveness analysis based on data derived from their c-RCT (Moore 2011), comparing 3-hourly repositioning using the 30° tilt overnight ($n = 99$, unadjusted for clustering) with standard care consisting of 6-hourly repositioning using the 90° lateral rotation overnight ($n = 114$, unadjusted for clustering), in participants recruited from 12 long-term aged-care facilities in Ireland. They compared the nursing time costs and incidence of pressure injury development over the four-week trial period. Nurse time was calculated from information recorded in the clinical study indicating number of turns per patient, nurses per turn, and nurse time per turn. A unit cost of EUR 23.94 per nurse hour was then applied, based on the rate for a staff nurse scale point 8 in mid-2009. Effectiveness was measured as pressure injury incidence (the primary outcome of the clinical trial), which would appear to be represented as the number of participants developing a new pressure injury during the four-week trial period. Moore 2013 also reported some data for the total cost of dressings for treating pressure injuries that developed during the trial but did not report a unit cost and did not include dressing costs in the incremental analysis. Moore 2013 estimated that implementing a 30° tilt 3-hourly regimen instead of a 90° tilt 6-hourly regimen (standard care) for the 588 (53.5%) individuals who were completely immobile or had very limited mobility across the 12 study sites would be associated with an annual cost difference favouring the intervention of EUR 512,800, equivalent to 21,462 hours of nurse time.

Incremental cost per event avoided

The 30° 3-hourly tilt positioning intervention was reported to be cost-saving in nurse time compared with standard care (mean nurse time cost per patient EUR 206.60 versus EUR 253.10, incremental difference EUR -46.50, 95% CI EUR -1.25 to EUR -74.60) (Moore 2013). The intervention was better than the control in terms of cost-effectiveness, since the trial also found the intervention to be more effective than the control. The lower nurse time cost for the

intervention group despite the greater repositioning frequency was due to the reduced time and reduced number of nurses required for each turn (Moore 2013).

Given that the intervention was better than the control, it was unnecessary for Moore 2013 to estimate an incremental cost-effectiveness ratio, although they did. There is some inconsistency in the reporting and interpretation of the incremental analysis made by Moore 2013, leading to a lack of clarity in the paper around the estimated cost-effectiveness. The authors of Moore 2013 suggest their efficacy outcome in the incremental analysis as both "patient free of PI" and "PI avoided". The rationale for alternating between outcome measures of "patient free of PI" and "PI avoided" is unclear. Nevertheless, in this instance, these outcome measures would appear to be equivalent, since the number of patients developing a pressure injury and the number of pressure injuries developing during the trial was the same ($n = 16$) (Moore 2011). Moore 2013 reported the incremental cost per patient free of pressure injury (EUR -73.40) and per pressure injury avoided (EUR -547.00). Although not explicitly stated, the estimated incremental cost-effectiveness ratios appear to be intended to represent an incremental cost per additional incremental outcome. However, these values are inconsistent with each other, given that the incidence of pressure injuries developing was the same in the trial (Moore 2011), regardless of whether it is defined as number of patients developing pressure injuries or number of pressure injuries developing during the trial. Furthermore, neither of these values could be confirmed from the data provided in the main body of the Moore 2013 paper. The former value (EUR -73.40 per patient free of PI) appears to have been incorrectly estimated from the data presented in the paper. The latter value of EUR -547 per (additional) pressure injury avoided is consistent with the efficacy data presented in the abstract, but the efficacy data presented in the abstract are inconsistent with efficacy data presented in the main body of the report, and do not precisely match the efficacy data provided in the original clinical trial report (Moore 2011). We did not undertake a GRADE assessment of this economic evaluation, so we cannot draw any firm or general conclusions regarding the relative costs or efficiency of 30° 3-hourly tilt overnight positioning compared with a 90° tilt 6-hourly regimen.

Despite this limitation in interpretation, the reported findings suggest that for every 100 people treated with the 3-hourly repositioning intervention rather than standard care, EUR 4650 would be saved in nurse time costs and an additional eight people would avoid a pressure injury. Moore 2013 concluded that repositioning every three hours using the 30° tilt is less costly in terms of nursing time and more effective than standard care involving repositioning every six hours using the 90° tilt.

The cost-minimisation analysis undertaken by Paulden 2014 explicitly assumes that the 2-, 3-, and 4-hourly repositioning strategies are equally effective in preventing pressure injury for nursing home residents at moderate to high risk of pressure injury who are using a high-density foam mattress, based on the findings of the Bergstrom 2013 trial. Their analysis therefore compares the costs of these strategies only.

Comparison 3: micromovement versus standard care

We included data from two trials with 498 participants in this comparison (Jin 2024; Zhang 2024). Both compared a

micromovement protocol to standard care on the development of intraoperative acquired pressure injury in people undergoing surgery. In the Jin 2024 trial, participants were in the supine position on the operating table, while in the Zhang 2024 study, participants were placed in the prone position.

Primary outcomes

Proportion of participants with a new pressure injury (any category/ stage 1 to 4)

Based on the similarity in interventions, we pooled the results of the two trials using the fixed-effect model ($I^2 = 0\%$). The occurrence of pressure injury may be lower with micromovement than standard care (pooled RR 0.28, 95% CI 0.11 to 0.67; 2 studies, 477 participants; low-certainty evidence; Analysis 9.1; [Summary of findings 9](#)). Our GRADE certainty of evidence assessment was low; we downgraded one level for high RoB and one level for imprecision (small sample size).

Secondary outcomes

Health-related quality of life

No trial reported health-related quality of life.

Procedural pain

No trial reported procedural pain.

Patient satisfaction

No trial reported patient satisfaction.

Costs

No trial reported economic costs.

Equity assessment

We did not assess health equity.

Reporting biases

As we included fewer than 10 studies in each meta-analysis, we did not generate funnel plots to evaluate the likelihood of publication bias.

DISCUSSION

See [Summary of findings 1](#); [Summary of findings 2](#); [Summary of findings 3](#); [Summary of findings 4](#); [Summary of findings 5](#); [Summary of findings 6](#); [Summary of findings 7](#); [Summary of findings 8](#); [Summary of findings 9](#)

Summary of main results

We included 11 clinical trials, with data from a total of 4462 participants receiving care in hospitals (general wards, intensive care units, operating theatres) and nursing homes. Participants' ages ranged from 18 to 90 years. The included studies compared standard care with repositioning interventions (i.e. two or more repositioning frequencies, different positioning regimens (30° tilt versus 90° lateral position; 30° versus 45° head of bed elevation), or micromovement protocols (supine versus prone)), and assessed the development of any category/stage pressure injury (1 to 4). In this review, we included five meta-analyses, four individual-trial analyses, and two cost-effectiveness analyses relating to included studies.

Repositioning regimens, including 2- and 3-hourly repositioning frequencies, are recommended in the pressure injury prevention clinical practice guidelines for individuals at prevention injury risk [4] and widely implemented clinically. In this 2026 update, the evidence supporting the effectiveness of these repositioning regimens on pressure injury prevention was uncertain. High heterogeneity was present in the pooled data of two meta-analyses and further subgroup analysis could not be explored, which limits confidence in the findings. Moreover, through our GRADE assessment [58], we judged the certainty of evidence as low or very low due to serious risk of bias related to lack of blinding (predominantly personnel) and imprecision due to small sample sizes/low number of events, missing data, or both. Furthermore, the routine implementation of 4-, 5-, or 6-hourly repositioning regimens is not recommended for individuals at pressure injury risk [4]. The cluster-RCT conducted in various nursing homes by Defloor 2005 showed that 46% of participants in the 4-hourly group and 62% of participants in the 6-hourly group developed a pressure injury, despite being nursed on viscoelastic foam mattresses, with the certainty of the evidence comparing these two frequencies assessed as very low (due to high risk of bias (lack of blinding, and missing data) and very serious imprecision). Consequently, we are unable to draw reliable conclusions regarding which repositioning regimens should be implemented.

The implementation of 30° or lower lateral positioning and 30° head of bed elevation is suggested in the pressure injury prevention clinical practice guidelines for individuals at pressure injury risk, along with avoiding 90° lateral positioning [4]. This 2026 update found very low certainty evidence that suggested no clear difference in the incidence of category/stage 1 or 2 pressure injury between 30° and 90° tilts. Furthermore, the considerable heterogeneity in the pooled data could not be further explored using subgroup analysis, so caution is required when interpreting our findings.

When regular repositioning regimens cannot be implemented in clinically unstable patients, it is good practice to initiate a short-term regimen of micromovement (i.e. small and incremental shifts in body position to reduce isolated tissue ischaemia by relieving pressure) and repositioning devices to offload pressure [4]. Although this approach makes physiological sense, the certainty of the evidence for its effectiveness is low. Furthermore, the implementation of micromovement is intended for short-term clinical use and not as a replacement for regular repositioning [4].

Regular repositioning, different positioning regimens, and the use of micromovement as an adjunct to regular repositioning are approaches used in clinical practice across all healthcare settings. The mostly low and very low certainty of evidence in this review does not permit us to reach reliable conclusions about the optimum position or frequency of repositioning, including micromovement. Our findings support the assertions of a scoping review undertaken in 2018 [111] and a recent systematic review [112]. Delivery of an individualised plan of offloading, coupled with other prevention strategies such as regular skin assessment and care, appropriate support surfaces [25, 113], and the involvement of patients in their care [4] are required.

Limitations of the evidence included in the review

We used the GRADE methodology to appraise our certainty about the body of evidence of the 11 clinical trials in relation to study

limitations (risk of bias), inconsistency of results, indirectness, imprecision, and publication bias, as specified in the *Cochrane Handbook for Systematic Reviews of Interventions* [58]. The GRADE appraisal is subjective in nature and requires each review author to make downgrading decisions using their judgement [58]. To ensure consistency in the GRADE appraisal, we specified predefined criteria (see RoB domains below), and used two independent appraisers, with a third review author used to reach consensus. As recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* [57], commentaries on economic evaluations included in the review do not require an assessment of their methodological quality, though we acknowledge this is a limitation of the review.

Limitations in study design and implementation

We assessed RoB according to five domains as well as overall: randomisation process, intervention deviations, missing outcome data, outcome measurement, and selection of reported results [44]. Our assessment of the RoB for several of these domains showed limitations in study design and implementation, which have been reported elsewhere in the review ([Supplementary material 5](#)). We had particular concerns about blinding of the intervention because this was difficult or impossible. Blinding of outcome assessors was a serious limitation in six trials (Defloor 2005; Ghezleji 2017; Moore 2011; Tarigan 2021; Zhang 2024; Zhou 2014). We judged the certainty of the evidence across most of the included trials as low or very low. We assessed the Pickham 2018 trial as providing moderate certainty of evidence.

The primary outcome, pressure injury development, requires a subjective clinical judgement regarding whether skin erythema or tissue damage has occurred (at least for category/stage 1 pressure injury), and only five of the 11 included studies used blinded outcome assessment along with a recognised classification system for pressure injury category/stage (Bergstrom 2013; Jin 2024; Manzano 2014; Pickham 2018; Young 2004) ([Supplementary material 8](#)).

Indirectness of evidence

There was no indirectness as the participants, interventions, and outcomes in the included studies were within the scope of the published review protocol.

Unexplained heterogeneity or inconsistency of results

Statistical heterogeneity ranged from none ($I^2 = 0\%$) to considerable ($I^2 = 75\%$). For pooled repositioning data with substantial heterogeneity (Analysis 2.1) or considerable heterogeneity (Analysis 3.1), we used the random-effects model to compute meta-analysis. In comparing the main outcome relative to tilt regimens, there was moderate statistical heterogeneity ($I^2 = 69\%$) (Analysis 6.1). This was likely due to the inconsistency in follow-up periods, which varied from 24 hours to 21 days, the lack of a definition of 'standard care', and the inconsistency with which interventions were delivered. In the presence of considerable heterogeneity, we did not conduct a subgroup analysis because this may yield misleading findings [66]. In comparing the main outcome relative to micromovement, we detected no heterogeneity (Analysis 9.1).

Imprecision of results

Imprecision was due to most studies being small and underpowered. The results of four analyses showed confidence

intervals that clearly crossed the line of no effect (Analysis 1.1; Analysis 2.1; Analysis 3.1; Analysis 6.1), indicating uncertainty about the relative effectiveness of the repositioning regimen or positioning for pressure injury prevention. While some of the studies included in this review were quite large in terms of the total number of participants (Bergstrom 2013; Pickham 2018), the numbers of events in these trials were lower than expected. This suggests that, to detect a difference between the interventions, larger samples may be required; however, this may be impractical. Furthermore, Ghezeljeh 2017 recruited a small sample of 120 participants (40 in the 30° head-of-bed elevation tilt group (intervention); 40 in the 45° head-of-bed elevation tilt group (intervention); and 40 in the control group), with zero pressure injury events in any of the groups.

Publication bias

We believe that our comprehensive electronic searches have identified all published RCTs addressing the review questions, thus limiting bias in the review process. That said, we identified nine unpublished ongoing studies, primarily through a search of clinical trial registries. After contacting the trial authors, we were advised that one ongoing study was finalised and the manuscript under review, while data analysis was underway in another ongoing study. Whilst the certainty of the body of evidence on repositioning regimens using various turning schedules, tilt angles, and micromovement is mostly low to very low, we did not downgrade the evidence for publication bias. Given the small number of studies in this review and the lack of results from unpublished studies, we were unable to assess publication bias using funnel plots.

Overall

Overall, there were weaknesses in the completeness and applicability of evidence reported in the included studies. Eight of the included trials were conducted in high-income countries (Defloor 2005; Jin 2024; Manzano 2014; Moore 2011; Young 2004; Zhang 2024). Eight of the 11 trials were conducted in acute care settings (wards, operating theatres, and intensive care units) (Ghezeljeh 2017; Jin 2024; Manzano 2014; Pickham 2018; Tarigan 2021; Young 2004; Zhang 2024; Zhou 2014). This may limit the generalisability of findings to lower-income and non-acute care (or non-hospital) settings.

The 11 studies included in this review had small sample sizes and few events, resulting in imprecise estimates that limit confidence in the treatment effects or lack thereof.

The primary study outcome reported in nine trials was the occurrence of pressure injuries. Two trials examined pressure injury as a secondary outcome (Ghezeljeh 2017; Zhou 2014). None of the included trials examined outcomes such as health-related quality of life, procedural pain, or patient satisfaction. Only two trial authors performed a parallel economic evaluation as a substudy (Moore 2013; Paulden 2014).

The focus of the interventions in the included studies varied. Three trials used tilts and 3-hourly overnight repositioning (Ghezeljeh 2017; Moore 2011; Young 2004), whilst one trial real-time wearable patient sensor data with and without visual reminders for repositioning (Pickham 2018). One trial used prone positioning (Zhou 2014). Four other trials used various repositioning frequencies and positions in combination with

different types of mattresses (Bergstrom 2013; Defloor 2005; Manzano 2014) or standard care (Tarigan 2021). Two trials used micromovement versus standard care in the operating theatre, with participants placed in either the supine (Jin 2024) or prone position (Zhang 2024).

Another limitation was inconsistency in follow-up periods, which included no follow-up (single assessment at the end of surgery) (Jin 2024; Zhang 2024), 24 hours (Manzano 2014; Young 2004), 3 days (Ghezeljeh 2017; Pickham 2018; Zhou 2014), 14 days (Tarigan 2021), 21 days (Bergstrom 2013), and 28 days (Defloor 2005; Moore 2011).

The variation in the types of support surfaces used in the clinical settings across the included trials limits generalisability. Inconsistencies in terminologies used to describe support surfaces added to this limitation. For instance, it is unclear whether a 'viscoelastic' mattress is similar to a 'high density foam' mattress. Furthermore, technological developments in mattress composition and materials, as well as bed design, have occurred over the 25 years during which these trials were undertaken, which also limits the external validity of these results, as other support surfaces are now in use.

Consumer engagement

No consumer engagement in the research process was evident in any of the included clinical trials.

Limitations of the review processes

We followed clearly defined, prespecified procedures to prevent potential bias in the review process. We conducted a comprehensive and systematic literature search that was both transparent and reproducible. Notwithstanding this, it is possible that we missed trials published in journals that were outside our search strategy.

Whilst we had intended to conduct a sensitivity analysis to test the robustness of the results for different assumptions about the outcomes of people who were lost to follow-up, we deemed this unnecessary due to the low volume and certainty of the evidence and our consequent inability to draw any conclusions (no sensitivity analysis would have helped in this regard).

Agreements and disagreements with other studies or reviews

Our results are consistent with others' assessment of the evidence for frequencies of repositioning and repositioning positions [111, 112]. The findings from a 2018 scoping review [111] and a 2023 systematic review [112] concluded that the evidence for optimal turning and repositioning schedules is inconclusive. Five of our trials (Bergstrom 2013; Defloor 2005; Manzano 2014; Moore 2011; Young 2004) were included in Asiri's 2023 systematic review [112], while only four trials (Bergstrom 2013; Defloor 2005; Manzano 2014; Moore 2011) were included in Chew's scoping review [111]. No additional evidence was located regarding micromovement as an adjunct to pressure injury prevention. A 2025 umbrella review of 14 systematic reviews synthesised the available evidence on economic evaluations of interventions for pressure injury prevention and treatment [19]. The authors stated there was insufficient evidence regarding the cost-effectiveness of repositioning, and recommended well-designed full economic evaluations be conducted as part of any future study [19].

AUTHORS' CONCLUSIONS

Implications for practice

Given the aetiology of pressure injury development is linked to deformation of the skin, blood, and lymphatic vessels, and underlying tissue, caused by prolonged mechanical loading [4], there are reasonable grounds to expect that repositioning hospitalised patients, and offloading pressure, would minimise the risk of deprivation of the oxygen and nutrients required for tissue health and repair. However, there is currently insufficient evidence to recommend one repositioning regimen in preference to another. Repositioning in some form is recommended in all clinical practice guidelines, but, to date, there is little evidence available from randomised controlled trials (RCTs) that addresses whether repositioning patients reduces the risk of pressure injury. The lack of evidence is a cause for concern, considering that estimates of the incidence of hospital-acquired pressure injuries range from less than 3% to over 48% of patients [15, 16, 17]. Individualised repositioning regimens should take into consideration the patient's clinical needs, preferences, comfort, level of activity, and their ability to reposition themselves [3, 4].

Equity-related implications for practice

We did not investigate equity-related characteristics in this review.

Implications for research

There is a compelling need for appropriately powered, high-quality, multicentre trials to evaluate the clinical effectiveness and cost-effectiveness of repositioning regimens for pressure injury prevention. The modest sample sizes in the trials included in this review is a major limitation. Larger numbers of participants are needed in future trials, particularly if cluster-randomised trials are conducted. Seven of the 11 trials included in the review were conducted in intensive care unit and operating theatre settings, where patients usually receive one-to-one care. Therefore, there is a need to conduct further research in acute care settings to address the rise in prevalence of hospital-acquired pressure injuries [3]. Consistency in the measures used to assess and classify pressure injuries (of any category/stage) is essential. Given the high costs associated with the prevention and treatment of pressure injuries, priority should be given to robust RCTs with economic evaluations.

Trialists should consider conducting trials that evaluate:

- repositioning frequencies and optimal positioning;
- use of manual repositioning regimens and electronic repositioning aids;
- effects of repositioning in high-risk patient populations (e.g. spinal cord injury);
- effects of position sensors on repositioning regimens;
- use of pressure sensor technologies to map pressure in relation to different tilt angles during repositioning;
- use of repositioning monitors to calculate/quantify patient repositioning whilst in bed;
- use of micromovement to augment regular repositioning regimens;
- economic costs (including incremental costs) of pressure injuries; and

- economic and social impacts of pressure injuries on patients' health-related quality of life (HRQoL) using valid and reliable measures.

Good-quality trials also need to address the methodological limitations identified in the trials included in this review. Trialists must ensure transparency of the research process and adhere to the CONSORT statement for reporting RCTs [114]. To minimise the sources of bias, trialists need to pay careful attention to elements of research design and execution with regard to the randomisation process, delivering interventions as intended, minimising missing outcome data, and ensuring robust outcome measurement [44]. If cluster-RCTs are used, trialists need to also consider the potential for bias in terms of selection bias, baseline comparability, analysis, and loss of clusters [64].

Equity-related implications for research

We did not investigate equity-related characteristics in this review.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD009958.pub4](https://doi.org/10.1002/14651858.CD009958.pub4).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Risk of bias

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 NPUAP-EPUAP pressure injury/ulcer classification system (2009, 2014)

Supplementary material 9 RoB 2 criteria

ADDITIONAL INFORMATION

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The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Professor Lisa Harvey, Professor of Rehabilitation Research, Faculty of Medicine and Health, The University of Sydney, Australia
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Gail Quinn, Cochrane Central Editorial Service

- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service
- Copy Editor (copy editing and production): Laura MacDonald, Cochrane Central Production Service
- Peer reviewers (provided comments and recommended an editorial decision): Sherry Reisner, Independent Health Writer (consumer reviewer); Lindsay Robertson, Cochrane (methods review); Jo Platt, Central Editorial information Specialist (search review)

The authors did not use artificial intelligence tools in the preparation of the written content of the manuscript.

Contributions of authors

We wish to acknowledge the contributions of the following authors and editors to the previous review and updates:

2026 author contributions

Sharon L Latimer: designed and co-ordinated the 2026 review update; ran the searches and edited the search methods sections; screened and reviewed studies for inclusion; extracted data; undertook and checked the quality assessment; performed the statistical analysis and checked the GRADE assessment; analysed and interpreted data; produced the first draft of the review update; contributed to writing or editing the review update; advised on the review update; approved the final review update prior to submission, and is a guarantor of the review.

Wendy P Chaboyer: conceived the original review; designed the review update; screened and reviewed studies for inclusion; analysed or interpreted data; contributed to writing or editing the review update; advised on the review update; performed previous work that was the foundation of the current review update; and approved the final review update prior to submission.

Sebastian Probst: designed the review update; extracted data; undertook quality assessment; analysed or interpreted data; contributed to writing or editing the review update; approved the final review update prior to submission.

Lukman Thalib: conceived the original review; designed the review update; analysed or interpreted data; performed and checked the statistical analysis; undertook the GRADE assessment; checked the quality of the statistical analysis; contributed to writing or editing the review update; advised on the review update; performed previous work that was the foundation of the current review update; approved the final review update prior to submission.

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Samuel Lapkin: designed the review update; extracted data; undertook quality assessment; analysed or interpreted data; contributed to writing or editing the review update; approved the final review update prior to submission.

Elizabeth McInnes: conceived the original review; designed the review update; analysed or interpreted data; checked the quality of data extraction and quality assessment; performed previous work

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Martin Downes: advised on the review update in regard to the economic analysis; contributed to writing or editing the review update; reviewed and updated the economic analysis; approved the final review update prior to submission.

Brigid M Gillespie: conceived the original review; led the 2014 review and 2020 update; designed the review update; screening adjudicator; extracted data; checked the quality of data extraction; analysed or interpreted data; undertook and checked the quality assessment; checked the statistical analysis and quality; contributed to writing or editing the review update; advised on the review update; performed previous work that was the foundation of the current review update; approved the final review update prior to submission; is a guarantor of the review update.

2020 author contributions

Brigid M Gillespie: conceived the review; designed the review update; co-ordinated the review update; extracted data; checked the quality of data extraction; analysed or interpreted data; undertook quality assessment; checked quality assessment; performed statistical analysis; checked the quality of the statistical analysis; produced the first draft of the review update; contributed to writing or editing the review update; advised on the review update; performed previous work that was the foundation of the current review update; performed economic analysis; approved the final review update prior to submission; is a guarantor of the review update.

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2014 author contributions

Brigid M Gillespie: conceived, designed and co-ordinated the review. Extracted data, undertook quality assessment, analysed or interpreted data, performed part of data analysis or interpretation, performed statistical analysis and completed the first draft of the review. Performed part of writing or editing, made an intellectual contribution to and approved final review prior to submission. Wrote to study authors, experts, and companies and acted as guarantor of the review.

Wendy P Chaboyer: conceived and designed the review, checked the quality of data extraction, analysed or interpreted data, performed part of data analysis or interpretation, performed part of the writing or editing, made an intellectual contribution to, secured funding for and approved the final version of the review prior to submission.

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Contributions of the Cochrane Wounds editorial base to previous versions

2020 update

Nicky Cullum (Co-ordinating Editor): edited the previous version of this review, advised on methodology, interpretation, and content; approved the final review prior to submission.

Gill Norman (Editor): edited this updated version of the review, advised on methodology, interpretation, and content; approved the final review prior to submission.

Gill Rizzello and Sally Bell-Seyer (Managing Editors): coordinated the editorial process; advised on interpretation and content and edited the review and the update.

Naomi Shaw and Sophie Bishop (Information Specialists): reviewed the search strategy.

Ursula Gonthier and Tom Patterson (Editorial Assistants): edited the Plain language summary and reference sections of the review.

2014 review

Nicky Cullum: advised on methodology, interpretation and protocol content, edited and re-wrote sections of the final review including reentering and analysing data, approved the final review prior to publication.

Sally Bell-Syer: co-ordinated the editorial process, advised on methodology, interpretation and content, edited the protocol.

Jo Dumville: checked the re-analysis of the data and checked the final version for publication after Nicky Cullum.

Ruth Foxlee: designed the search strategy and edited the search methods section.

Declarations of interest

None of the review authors is an author on any of the included studies.

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Registration and protocol

This 2026 update of the review followed the protocol developed and published in 2012: Gillespie BM, Chaboyer WP, McInnes E, Kent B, Whitty JA. Repositioning for pressure ulcer prevention in adults (Protocol). Cochrane Database of Systematic Reviews 2012, Issue 7. Art. No.: CD009958. DOI: 10.1002/14651858.CD009958.

What's new

Date	Event	Description
4 June 2026	New search has been performed	Updated search run for this second update of the review
4 June 2026	New citation required but conclusions have not changed	Three new studies included but conclusions unchanged

History

Protocol first published: Issue 7, 2012

Review first published: Issue 4, 2014

Date	Event	Description
28 May 2020	New search has been performed	First update. New search with five new studies and one economic substudy added.
28 May 2020	New citation required but conclusions have not changed	Conclusions unchanged
24 February 2015	Amended	Contact details updated

Data, code and other materials

As part of the published Cochrane Review, the following are made available for download for users of the Cochrane Library: full search strategies for each database; full citations of each study included, ongoing, or excluded at the full-text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; analysis data, including overall estimates and settings, subgroup estimates, and individual data rows. Appropriate permissions have been obtained for such use.

Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods.

The data package is available at [Supplementary material 7](#).

REFERENCES

1. National Pressure Ulcer Advisory Panel. Pressure ulcer staging system. Available from <https://www.npuap.org> 2007.
2. National Pressure Ulcer Advisory Panel (NPUAP). National Pressure Ulcer Advisory Panel (NPUAP) announces a change in terminology from pressure ulcer to pressure injury and updates the stages of pressure injury. Available from <https://www.npuap.org/national-pressure-ulcer-advisory-panel-npuap-announces-a-change-in-terminology-from-pressure-ulcer-to-pressure-injury-and-updates-the-stages-of-pressure-injury/> (accessed 15 October 2018).
3. European Pressure Ulcer Advisory Panel, National Pressure Injury Advisory Panel and Pan Pacific Pressure Injury Alliance; Emily Haesler (Ed). Prevention and treatment of pressure ulcers/injuries: clinical practice guideline. Available from <https://www.internationalguideline.com> 2019; **Third edition**:1-405.
4. National Pressure Injury Advisory Panel, European Pressure Ulcer Advisory Panel and Pan Pacific Pressure Injury Alliance; Emily Haesler (Ed). Prevention and treatment of pressure ulcers/injuries: clinical practice guideline. The International Guideline. Fourth edition (online). Available from <https://internationalguideline.com/the-international-guideline> 2025.
5. National Pressure Ulcer Advisory Panel (NPUAP), European Pressure Ulcer Advisory Panel (EPUAP), Pan Pacific Pressure Injury Alliance (PPPIA). Prevention and treatment of pressure ulcers: quick reference guide. Available from <https://www.npuap.org/wp-content/uploads/2014/08/Updated-10-16-14-Quick-Reference-Guide-DIGITAL-NPUAP-EPUAP-PPPIA-16Oct2014.pdf> (accessed 15 October 2018).
6. Gefen A, Brienza DM, Cuddigan J, Haesler E, Kottner J. Our contemporary understanding of the aetiology of pressure ulcers/pressure injuries. *International Wound Journal* 2022;**19**(3):692-704. [DOI: [10.1111/iwj.13667](https://doi.org/10.1111/iwj.13667)]
7. World Health Organization. International Classification of Diseases Eleventh Revision (ICD-11). Available from <https://icd.who.int/browse/2024-01/mms/en#455330172> 2022; **License: CC BY-ND 3.0 IGO**.
8. European Pressure Ulcer Advisory Panel (EPUAP), National Pressure Ulcer Advisory Panel (NPUAP). Prevention and treatment of pressure ulcers: quick reference guide. https://www.epuap.org/wp-content/uploads/2016/10/final_quick_prevention.pdf (accessed 15 October 2018).
9. Liu S, Rawson H, Islam R, Team V. Impact of pressure injuries on health-related quality of life: A systematic review. *Wound Repair and Regeneration* 2024;**33**(1):e13236. [DOI: [10.1111/wrr.13236](https://doi.org/10.1111/wrr.13236)]
10. Padula W, Delarmente B. The national cost of hospital-acquired pressure injuries in the United States. *International Wound Journal* 2019;**16**(3):634-40. [DOI: [10.1111/iwj.13071](https://doi.org/10.1111/iwj.13071)]
11. World Health Organization. Global patient safety action plan 2021–2030: towards eliminating avoidable harm in health care. <https://www.who.int/teams/integrated-health-services/patient-safety/policy/global-patient-safety-action-plan-2021>; **License: CC BY-NC-SA 3.0 IGO**:i-108.
12. Sugathapala RDUP, Latimer S, Balasuriya A, Chaboyer W, Gillespie MB. Prevalence and incidence of pressure injuries among older people living in nursing homes: a systematic review and meta-analysis. *International Journal of Nursing Studies* 2023;**148**:104605.
13. Australian Commission on Safety and Quality in Health Care. Hospital-acquired complications (HACs). <https://www.safetyandquality.gov.au/our-work/indicators-measurement-and-reporting/hospital-acquired-complications-hacs> 2022.
14. Duke G, Moran J, Bersten A, Bihari S, Roodenburg O, Karnon J, Hirth S, Hakendorf P, Santamaria J. Hospital-acquired complications: the relative importance of hospital- and patient-related factors. *Medical Journal of Australia* 2022;**216**(5):242-47. [DOI: [10.5694/mja2.51375](https://doi.org/10.5694/mja2.51375)]
15. Chaboyer W, Bucknall T, Webster J, McInnes E, Gillespie MB, Banks M, et al. The effect of a patient centred care bundle intervention on pressure ulcer incidence (INTACT): a cluster randomised trial. *International Journal of Nursing Studies* 2016;**64**:63-71.
16. Mulligan S, Prentice J, Scott L. WoundsWest Wound Prevalence Survey 2011 State-wide Overview Report. Ambulatory Care Services, Department of Health: Perth, Western Australia; 2011 1-41.
17. Li Z, Lin F, Thalib L, Chaboyer W. Global prevalence and incidence of pressure injuries in hospitalised adult patients: a systematic review and meta-analysis. *International Journal of Nursing Studies* 2020;**105**:103546. [DOI: [10.1016/j.ijnurstu.2020.103546](https://doi.org/10.1016/j.ijnurstu.2020.103546)]
18. Nghiem S, Campbell J, Walker R, Byrnes J, Chaboyer W. Pressure injuries in Australian public hospitals: a cost of illness study. *International Journal of Nursing Studies*, 2022;**130**:104191. [DOI: [10.1016/j.ijnurstu.2022.104191](https://doi.org/10.1016/j.ijnurstu.2022.104191)]
19. Klugar M, Klugarová J, Hussain S, Vrbová T, Slezáková S, Bůřilová P, et al. Economic evaluations of interventions for the prevention and treatment of pressure ulcers: an umbrella review. *JBI Evidence Synthesis* 2023;**23**(8):1536-64. [DOI: [10.11124/JBIES-23-00488](https://doi.org/10.11124/JBIES-23-00488)]
20. Lindgren M, Unosson M, Fredrikson M, Ek AC. Immobility - a major risk factor for development of pressure ulcers among adult hospitalised patients: a prospective study. *Scandinavian Journal of Caring Sciences* 2004;**18**(1):57-63.
21. Chung M, Widdel M, Kirchhoff J, Sellin J, Jelali M, Geiser F, Mücke M, Conrad R. Risk factors for pressure injuries in adult patients: a narrative synthesis. *International Journal of Environmental Research and Public Health* 2022;**19**(2):761. [DOI: [10.3390/ijerph19020761](https://doi.org/10.3390/ijerph19020761)]

22. Braden BJ, Maklebust J. Preventing pressure ulcers with the Braden scale: an update on this easy-to-use tool that assesses a patient's risk. *American Journal of Nursing* 2005;**105**(5):70-2.
23. Moore Z, Patton D. Risk assessment tools for the prevention of pressure ulcers (Review). *Cochrane Database of Systematic Reviews* 2019, Issue 1. Art. No: CD006471. [DOI: [10.1002/14651858.CD006471.pub4](https://doi.org/10.1002/14651858.CD006471.pub4)]
24. Patton D, Moore Z, Boland F, Chaboyer W, Latimer S, Walker R, Avsar P. Dressings and topical agents for preventing pressure ulcers (Review). *Cochrane Database of Systematic Reviews* 2024, Issue 12. Art. No: CD009362. [DOI: [10.1002/14651858.CD009362.pub4](https://doi.org/10.1002/14651858.CD009362.pub4)]
25. McInnes E, Jammali-Blasi A, Bell-Syer SE, Dumville JC, Middleton V, Cullum N. Support surfaces for pressure ulcer prevention. *Cochrane Database of Systematic Reviews* 2015, Issue 9. Art. No: CD001735. [DOI: [10.1002/14651858.CD001735.pub5](https://doi.org/10.1002/14651858.CD001735.pub5)]
26. Kottner J, Coleman S. The theory and practice of pressure ulcer/injury risk assessment: a critical discussion. *Journal of Wound Care* 2023;**32**(9):560-69. [DOI: [10.12968/jowc.2023.32.9.560](https://doi.org/10.12968/jowc.2023.32.9.560)]
27. Exton-Smith AN, Sherwin RW. The prevention of pressure sores: significance of spontaneous bodily movements. *Lancet* 1961;**278**(7212):1124-6.
28. Sonenblum S, Sprigle S. The impact of tilting on blood flow and localized tissue loading. *Journal of Tissue Viability* 2011;**20**(1):3-13. [DOI: [10.1016/j.jtv.2010.10.001](https://doi.org/10.1016/j.jtv.2010.10.001)]
29. Langemo D, Anderson J, Hanson D, Thompson P, Johnson E. The conundrum of turning/repositioning frequency, sleep surface selection, and sleep disruption in preventing pressure injury in healthcare settings. *Advances in Skin & Wound Care* 2022;**35**(5):252-59.
30. European Pressure Ulcer Advisory Panel Guideline (EPUAP). A policy statement on the prevention of pressure ulcers from the European Pressure Ulcer Advisory Panel. *British Journal of Nursing* 1998;**7**:888-90.
31. Norton D, McLaren R, Exton-Smith AN. An Investigation of Geriatric Nursing Problems in Hospital. New York (NY): Churchill Livingstone, 1962.
32. Palmen NG. Prevention of Pressure Sores in a General Hospital: a Study into the Effect of a Nursing Intervention Focusing on Pressure Sore Prevention [Masters thesis]. Maastricht, the Netherlands: Maastricht University, 1987.
33. Smith AM, Malone JA. Preventing pressure ulcers in institutionalised elders: assessing the effects of small, unscheduled shifts in body position. *Decubitus* 1990;**3**(4):20-4.
34. Australian Institute of Health and Welfare. Australian hospital statistics 2007-08. www.aihw.gov.au/reports/hospitals/ahs-2007-08/contents/table-of-contents.
35. United States Department of Labor. Case and demographic characteristics for work-related injuries and illnesses involving days away from work. Resource table 10: detailed occupation by selected parts of body affected. <https://www.bls.gov/iif/oshcdnew2016.htm>.
36. Carskadon MA, Dement WC. Normal human sleep: an overview. In: Kryger MH, Roth T, Dement WC, editor(s). *Principles and Practice of Sleep Medicine*. 4th edition. Philadelphia (PA): Elsevier Sanders, 2005:13-23.
37. Dawson A, McLennan S, Schiller S, Jull GA, Hodges PW, Stewart S. Interventions to prevent back pain and back injury in nurses: a systematic review. *Occupational and Environmental Medicine* 2007;**64**:642-50.
38. Humphries J. Sleep disruption in hospitalised adults. *Medical-Surgical Nursing* 2008;**17**(6):391-5.
39. Raymond I, Ancoli-Israel S, Choiniere M. Sleep disturbances, pain and analgesia in adults hospitalised for burn injuries. *Sleep Medicine* 2004;**5**(6):551-9.
40. Vieira E, Kumar S. Safety analysis of patient transfers and handling tasks. *Quality and Safety in Health Care* 2009;**18**(5):380-4.
41. Trinkoff A, Storr C, Lipscomb J. Physically demanding work and inadequate sleep, pain, medication use and absenteeism in registered nurses. *Journal of Occupational and Environmental Medicine* 2001;**43**(4):355-63.
42. US Bureau of Labor Statistics. Injuries, illnesses, and fatalities: Registered nurses, 2016-2020. <https://www.bls.gov/iif/snapshots/osn-registered-nurses-2016-20.htm> (last updated February 11 2022) 2022.
43. Safe Work Australia. Workers' compensation interactive data. <https://data.safeworkaustralia.gov.au/interactive-data/topic/workers-compensation> (accessed August 8 2025) 2024.
44. Sterne JAC, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;**366**:14898. [DOI: <https://doi.org/10.1136/bmj.l4898>]
45. Higgins JP, Sterne JA, Savović J, Page MJ, Hróbjartsson A, Boutron I, et al. A revised tool for assessing risk of bias in randomized trials. *Cochrane Database of Systematic Reviews* 2016;**10**(Suppl 1):29-31.
46. Staniszewska S, Brett J, Simera I, Seers K, Mockford C, Goodlad S, et al. GRIPP2 reporting checklists: tools to improve reporting of patient and public involvement in research. *Research Involvement and Engagement* 2017;**3**(13):1-11. [DOI: <https://doi.org/10.1186/s40900-017-0062-2>]
47. Gillespie BM, Walker RM, Latimer SL, Thalib L, Whitty JA, McInnes E, Chaboyer WP. Repositioning for pressure injury prevention in adults. *Cochrane Database of Systematic Reviews* 2020, Issue 6. Art. No: CD009958. [DOI: [10.1002/14651858.CD009958.pub3](https://doi.org/10.1002/14651858.CD009958.pub3)]
48. Endnote version 20.3. The Endnote Team. Philadelphia, PA: Clarivate, 2020. Available at <https://www.endnote.com>.

49. Covidence. Veritas Health Innovation, Melbourne, Australia, Version accessed 13 May 2025. Melbourne, Australia: Veritas Health Innovation, 2025. Available at <https://www.covidence.org>.
50. Lefebvre C, Manheimer E, Glanville J. Chapter 6: Searching for studies. In: Higgins JP, Green S, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0 (updated March 2011). The Cochrane Collaboration, 2011. Available from handbook.cochrane.org.
51. Scottish Intercollegiate Guidelines Network. Search Filters. <https://www.sign.ac.uk/search-filters.html> (accessed 15 October 2018).
52. Centre for Reviews and Dissemination. NHS EED search filters. <https://www.crd.york.ac.uk/crdweb/searchstrategies.asp> (accessed prior to 20 February 2020).
53. Gillespie BM, Chaboyer WP, McInnes E, Kent B, Whitty JA, Thalib L. Repositioning for pressure ulcer prevention in adults. *Cochrane Database of Systematic Reviews* 2014, Issue 4. Art. No: CD009958. [DOI: [10.1002/14651858.CD009958.pub2](https://doi.org/10.1002/14651858.CD009958.pub2)]
54. Gillespie BM, Chaboyer WP, McInnes E, Kent B, Whitty JA. Repositioning for pressure ulcer prevention in adults. *Cochrane Database of Systematic Reviews* 2012, Issue 7. Art. No: CD009958. [DOI: [10.1002/14651858.CD009958](https://doi.org/10.1002/14651858.CD009958)]
55. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5 (updated August 2024). Cochrane 2024. Available from <https://www.cochrane.org/handbook>.
56. Review Manager (RevMan). Version 9.8.2. Copenhagen: The Cochrane Collaboration (August 08, 2025), 2025. Available at <https://revman.cochrane.org>.
57. Aluko P, Graybill E, Craig D, Henderson C, Drummond M, Wilson ECF, et al; on behalf of the Campbell and Cochrane Economics Methods Group. Chapter 20: Economic evidence [last updated August 2023]. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5 (updated August 2024). Available from <https://www.cochrane.org/handbook>.
58. Schünemann HJ, Higgins JPT, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence (updated August 2023). In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5. Available from cochrane.org/handbook. Cochrane, 2024.
59. Deeks J. Issues in the selection of a summary statistic for meta analysis of clinical trials with binary outcomes. *Statistics in Medicine* 2002;**21**(1):575-600.
60. Husereau D, Drummond M, Petrou S, Carswell C, Moher D, Greenberg D, et al. Consolidated Health Economic Evaluation Reporting Standards (CHEERS) - explanation and elaboration: a report of the ISPOR health economic evaluation publication guidelines good reporting practices task force. *Value in Health* 2013;**16**(2):231-50.
61. Shemilt I, Thomas J, Morciano M. A web-based tool for adjusting costs to a specific target currency and price year. *Evidence and Policy* 2010;**6**(1):51-9.
62. Brazier J, Ratcliffe J, Saloman J, Tsuchiya A. Measuring and Valuing Health Benefits for Economic Evaluation. New York, USA: Oxford University Press, 2017.
63. Johnston BC, Patrick DL, Devji T, Maxwell LJ, Bingham III CO, Beaton D, et al. Chapter 18: Patient-reported outcomes [last updated October 2019]. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5. Available from <https://www.cochrane.org/handbook>. Cochrane, 2024.
64. Higgins JP, Eldridge S, Li T. Chapter 23: Including variants on randomized trials (last updated October 2019). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5. Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
65. Page MJ, Higgins JPT, Sterne JAC. Chapter 13: Assessing risk of bias due to missing evidence in a meta-analysis (updated August 2024). In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5. Available from cochrane.org/handbook. Cochrane, 2024.
66. Deeks JJ, Higgins JPT, Altman DG, McKenzie JE, Veroniki AA. Chapter 10: Analysing data and undertaking meta-analyses (updated November 2024). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al. *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5. Available from <https://www.cochrane.org/handbook> 2024.
67. GRADEpro GDT. Version 14 August 2025. Hamilton (ON): McMaster University (developed by Evidence Prime), 2025. Available at <https://gradepro.org>.
68. Ryan R, Hill S. How to GRADE the quality of the evidence. Cochrane Consumers and Communication Group. Version 3.0. <https://cccr.org.cochrane.org/author-resources> (accessed prior to 20 February 2020).
69. Schünemann HJ, Tugwell P, Reeves BC, Akl EA, Santesso N, Spencer FA, et al. Non-randomized studies as a source of complementary, sequential or replacement evidence for randomized controlled trials in systematic reviews on the effects of interventions. *Research Synthesis Methods* 2013;**4**:49-62.
70. Cochrane. Cochrane framework for engagement and involvement of patients, carers, and the public. <https://www.cochrane.org/about-us/news/cochrane-framework-engagement-and-involvement-patients-carers-and-public> 2022.
71. Zhou X, Dawei L, Zhang G, Cui N, He H, Zhao H, et al. The effects of prone position ventilation combined with recruitment maneuvers on outcomes in patients with severe

acute respiratory distress syndrome. *Chung-Hua Nei Ko Tsa Chih [Chinese Journal of Internal Medicine]* 2014;**53**(6):437-41. [DOI: <https://doi.org/10.3760/cma.j.issn.0578-1426.2014.06.007>]

72. Jin Y, Liu J, Shentu Y, Xuan F, Guo H, Li Y. Effect of "micromovement" in preventing intraoperative acquired pressure injuries among patients undergoing surgery in supine position. *International Wound Journal* 2024;**21**(2):e14408. [DOI: <https://doi.org/10.1111/iwj.14408>]

73. Tarigan S, Saldy Y, Yuliana S. Effect of interface pressure and skin surface temperature on pressure injury incidence: a turning schedule pilot study. *Journal of Wound Care* 2021;**30**(8):632-41.

74. Zhang J, Wang P, Qi L, Feng S, Zhang F. The effect of micro-movement on prevention of intraoperative acquired pressure injury in overweight patients undergoing posterior lumbar surgery: a randomized controlled trial. *Journal of Orthopaedic Surgery and Research* 2024;**19**:823:823-33. [DOI: [10.1186/s13018-024-05312-8](https://doi.org/10.1186/s13018-024-05312-8)]

75. Bergstrom N, Horn SD, Rapp MP, Stern A, Barrett R, Watkiss M. Turning for Ulcer ReductionN: a multisite randomised clinical trial in nursing homes. *Journal of the American Geriatrics Society* 2013;**61**(20):1705-13. [DOI: [10.1111/jgs.12440](https://doi.org/10.1111/jgs.12440)]

76. Defloor T, De Bacquer D, Grypdonck MH. The effect of various combinations of turning and pressure reducing devices on the incidence of pressure ulcers. *International Journal of Nursing Studies* 2005;**42**(1):37-46. [DOI: <https://doi.org/10.1016/j.ijnurstu.2004.05.013>]

77. Ghezleji T, Kalhor L, Moghadam O, Lahiji M, Haghani H. The comparison of the effect of the head of bed elevation to 30 and 45 degrees on the incidence of ventilator associated pneumonia and the risk for pressure ulcers: a controlled randomised clinical trial. *Iranian Red Crescent Medical Journal* 2017;**19**(7):e14224. [DOI: <https://doi.org/10.5812/ircmj.14224>]

78. Manzano F, Colmenero M, Pérez-Pérez AM, Roldán D, del Mar Jiménez-Quintana M, Mañas MR, et al. Comparison of two repositioning schedules for the prevention of pressure ulcers in patients on mechanical ventilation with alternating pressure air mattresses. *Intensive Care Medicine* 2014;**40**(11):1679-87. [DOI: <https://doi.org/10.1007/s00134-014-3424-3>]

79. Moore Z, Cowman S, Conroy RM. A randomised controlled clinical trial of repositioning, using the 30° tilt, for the prevention of pressure ulcers. *Journal of Clinical Nursing* 2011;**20**(17-18):2633-44. [DOI: <https://doi.org/10.1111/j.1365-2702.2011.03736.x>]

80. Pickham D, Berte N, Pihulic M, Valdex A, Mayer B, Sesai M. Effect of a wearable patient sensor on care delivery for preventing pressure injuries in acutely ill adults: a pragmatic randomised clinical trial (LS-HAPI study). *International Journal of Nursing Studies* 2018;**80**:12-9. [DOI: <https://doi.org/10.1016/j.ijnurstu.2017.12.012>]

81. Young T. The 30 degree tilt position vs the 90 degree lateral and supine positions in reducing the incidence of non-blanching erythema in a hospital inpatient population: a randomised controlled trial. *Journal of Tissue Viability*

2004;**14**(3):88, 90, 92-6. [DOI: [https://doi.org/10.1016/S0965-206X\(04\)43004-6](https://doi.org/10.1016/S0965-206X(04)43004-6)]

82. ChiCTR2100042476. Effects of different degrees of backrest elevation on ventilator-associated pneumonia and pressure ulcer in ICU patients with mechanical ventilation. <https://trialsearch.who.int/Trial2.aspx?TrialID=ChiCTR2100042476> (first received January 22, 2021). [WEBSITE: <https://trialsearch.who.int/Trial2.aspx?TrialID=ChiCTR2100042476>]

83. ChiCTR2100046245. Application value of 30 degree reclining position in preventing aspiration and pressure sores in neurosurgical patients. trialsearch.who.int/Trial2.aspx?TrialID=ChiCTR2100046245 (first received May 12 2021). [WEBSITE: trialsearch.who.int/Trial2.aspx?TrialID=ChiCTR2100046245]

84. KCT0009366. Effects of pressureinjury prevention nursing using real-time pressure mapping. trialsearch.who.int/Trial2.aspx?TrialID=KCT0009366 (first received April 29 2024). [WEBSITE: trialsearch.who.int/Trial2.aspx?TrialID=KCT0009366]

85. NCT04550182. Evaluation of the impact of the repositioning schedule adapted to the risk of pressure ulcer of patients in intensive care unit (PROMESREA). <https://clinicaltrials.gov/study/NCT04550182> (first received September 16 2020). [WEBSITE: <https://clinicaltrials.gov/study/NCT04550182>]

86. NCT04604665. Evaluation of two levels of frequency of repositioning in the reduction of pressure ulcers. <https://clinicaltrials.gov/study/NCT04604665> (first received October 27th 2020). [WEBSITE: <https://clinicaltrials.gov/study/NCT04604665>]

87. NCT05549830. Effect of different positioning before, during and after surgery on pressure injury. <https://clinconnect.io/trials/NCT05549830> (first received September 22, 2022). [DOI: <https://doi.org/10.18203/2349-3259.ijct20240042>]

88. NCT02479425. Two points versus three points turning in prevention of bed sores in critically ill patients. clinicaltrials.gov/ct2/show/NCT02479425 (first received 24 June 2015). [WEBSITE: clinicaltrials.gov/ct2/show/NCT02479425]

89. NCT03048357. Effectiveness of freedom bed compared to manual turning in prevention of pressure injuries in persons with limited mobility due to traumatic brain injury and/or spinal cord injury. clinicaltrials.gov/ct2/show/NCT03048357 (first received 9 February 2016). [WEBSITE: clinicaltrials.gov/ct2/show/NCT03048357]

90. NCT03454230. Pilot study for evaluation of the procedure impact of mobilization adapted to pressure ulcer risk for patients in intensive care (PROMES-REA PILOT). clinicaltrials.gov/ct2/show/NCT03454230 (first received 5 March 2018). [WEBSITE: clinicaltrials.gov/ct2/show/NCT03454230]

91. Moore Z, Cowman S, Posnett J. An economic analysis of repositioning for the prevention of pressure ulcers. *Journal of Clinical Nursing* 2013;**22**(15-16):2354-60.

92. Paulden M, Bergstrom N, Horn SD, Rapp MP, Barrett R, Watkiss M, et al. Turning for Ulcer Reduction (TURN) study: an

economic analysis. Ontario Health Technology Assessment Series 2014;**14**(12):1-24.

- 93.** Cumming RG, Sherrington C, Lord SR, Simpson JM, Vogler C, Cameron ID, et al; Prevention of Older People's Injury Falls Prevention in Hospitals Research Group. Cluster randomised trial of a targeted multifactorial intervention to prevent falls among older people in hospital. *BMJ* 2008;**336**(7547):758-60.
- 94.** Beuret P, Carton MJ, Nourdine K, Kaaki M, Tramoni G, Ducreaux JC. Prone position as prevention of lung injury in comatose patients: a prospective, randomised, controlled study. *Intensive Care Medicine* 2002;**28**:564-9.
- 95.** Girard R, Baboi L, Ayzac L, Richard JC, Guérin C; Proseva trial group. The impact of patient positioning on pressure ulcers inpatients with severe ARDS: results from a multicentre randomised controlled trial on prone positioning. *Intensive Care Medicine* 2014;**40**(3):397-403.
- 96.** Su MY, Lin SQ, Zhou YW, Liu SY, Lin A. A prospective, randomised, controlled study of a suspension positioning system used with elderly bedridden patients with neurogenic fecal incontinence. *Ostomy Wound Management* 2015;**61**(1):30-9.
- 97.** Taccone P, Pesenti A, Latini R, Polli F, Vagginelli F, Mietto C, et al; Prone-Supine II Study Group. Prone positioning in patients with moderate and severe acute respiratory distress syndrome: a randomised controlled trial. *JAMA* 2009;**18**:1977-84.
- 98.** Vanderwee K, Grypdonck MH, De Bacquer D, Defloor T. Effectiveness of turning with unequal time intervals on the incidence of pressure ulcer lesions. *Journal of Advanced Nursing* 2007;**57**(1):59-68.
- 99.** Voggenreiter G, Aufmkolk M, Stiletto RJ, Baacke MG, Waydhas C, Ose C, et al. Prone positioning improves oxygenation in post-traumatic lung injury: a prospective randomised trial. *Journal of Trauma, Injury, Infection, and Critical Care* 2005;**59**:333-43.
- 100.** De Meyer D, Van Hecke A, Verhaeghe S, Beeckman D. PROTECT-Trial: A cluster RCT to study the effectiveness of a repositioning aid and tailored repositioning to increase repositioning compliance. *Journal of Advanced Nursing* 14 December 2018;**75**(5):1085-98.
- 101.** Jiang Q, Liu Y, Yu H, Song S, Li G, Liu H, Zhou Y, Zhu Y, Jia J, Huang Y, Wang J. A multicenter, comparative study of two pressure-redistribution mattresses with repositioning intervals for critical care patients. *Advances in Skin & Wound Care* July 3, 2019;**33**(3):1-9.
- 102.** Kapp S, Gerdtz M, Gefen A, Padula W, Alves P, Trevellini C, Ghosh A, Shea A, Cross A, Sousa I, Santamaria N. Clinical and cost effectiveness of a system for turning and positioning intensive care unit patients, when compared to usual care turning and positioning devices, for the prevention of hospital-acquired pressure injuries. A randomised controlled trial. *International Wound Journal* 09 June 2023;**20**(9):3567-79.

- 103.** Malekara N, Larti N, Shirzadi A, Bozorgmehr R. Comparing the effects of curvilinear position and micromovement on prevention of intraoperative acquired pressure injuries among patients undergoing surgery in the supine position: A randomized controlled trial. *Injury* 7 April 2025;**56**(6):112327.
- 104.** Özkan E, Turhan, K. The effect of Position time on the formation of Pressure Ulcer in Bedridden Patients [Yatağa Bağımlı Hastalarda Pozisyon Süresinin Basınç Yarısı Oluşumu Üzerine Etkisi]. *Journal of Nursing Sciences* 2019;**11**(3):246-54.
- 105.** Yap T, Horn S, Sharkey D, Zheng T, Bergstrom N, Colon-Emeric C, Sabol V, Alderden J, Yap W, Kennerly S. Effect of varying repositioning frequency on pressure injury prevention in nursing home residents: TEAM-UP trial results. *Advances in Skin & Wound Care* 23 December, 2021;**35**(6):315-25.
- 106.** Ziegler S, Schmoor C, Schöler L, Schepputat S, Takem E, Grotejohann B, Steinbrenner I, Feuchtinger J. Potential for reducing immobility times of a mobility monitor in-bed sensor system - a stepped-wedge cluster-randomised trial. *BMC Nursing* 16 December 2023;**22**(1):478-88.
- 107.** Kang S, DiStefano M, Yehia F, Koszalka M, Padula W. Critical care beds with continuous lateral rotation therapy to prevent ventilator-associated pneumonia and hospital-acquired pressure injury: a cost-effectiveness analysis. *Journal of Patient Safety* 2021;**17**(2):149-55.
- 108.** Nherera L, Larson B, Cooley A, Reinhard P. An economic analysis of a wearable patient sensor for preventing hospital-acquired pressure injuries among the acutely ill patients. *International Journal of Health Economics and Management* 09 April, 2021;**21**(4):457-71.
- 109.** Padula W, Crawford S, Kennerly S, Yap T. Estimating the value of repositioning timing to streamline pressure injury prevention efforts in nursing homes: A cost-effectiveness analysis of the 'TEAM-UP' clinical trial. *International Wound Journal* 1 November 2023;**21**(3):e14452.
- 110.** Pickham D, Ballew B, Ebong K, Shinn J, Lough ME, Mayer B. Evaluating optimal patient-turning procedures for reducing hospital-acquired pressure ulcers (LS-HAPU): study protocol for a randomized controlled trial. *Trials* 2016;**6**(17):190. [DOI: [10.1186/s13063-016-1313-5](https://doi.org/10.1186/s13063-016-1313-5)]
- 111.** Chew H-S J, Thiara E, Lopez V, Shorey S. Turning frequency in adult bedridden patients to prevent hospital-acquired pressure ulcer: a scoping review. *International Wounds Journal* 2018;**15**(2):225-36. [DOI: [10.1111/iwj.12855](https://doi.org/10.1111/iwj.12855)]
- 112.** Asiri S. Turning and repositioning frequency to prevent hospital-acquired pressure injuries among adult patients: Systematic review. *INQUIRY* 2023;**60**:1-10. [DOI: [10.1177/00469580231215209](https://doi.org/10.1177/00469580231215209)]
- 113.** Shi C, Dumville J, Cullum N, Rhodes S, McInnes E. Alternative reactive support surfaces (non-foam and non-air-filled) for preventing pressure ulcers. *Cochrane Database of Systematic Reviews* 2021, Issue 5. Art. No: CD013623. [DOI: [10.1002/14651858.CD013623.pub2](https://doi.org/10.1002/14651858.CD013623.pub2)]

114. CONSORT Group. The CONSORT Statement. Consolidated Standards of Reporting Trials. <https://www.consort-statement.org> (accessed 15 October 2018).

ADDITIONAL TABLES

Table 1. History of differences between protocol and review versions

Version	Change
Changes in 2014 version	We had planned in the protocol to undertake subgroup analyses based on type of setting (long-term and acute care) and the type of patient. Although one study was conducted in an acute care setting, the others were set in long-term care facilities, and all with geriatric patients. Instead, we undertook subgroup analysis with regard to tilt regimes (i.e. 30° versus 90° tilt) in relation to participants who developed a grade 1 pressure ulcer.
Changes in the 2020 update	- We added the outcome 'patient satisfaction' to the summary of findings tables. - We changed our inclusion criteria to exclude studies if incidence of pressure ulcers were not included as a primary or secondary outcome because this outcome is central to the purpose of the review. - We updated our search terms by adding new intervention names and relevant database indexing terms. - We changed the term 'pressure ulcer' to 'pressure injury', in keeping with the National Pressure Ulcer Advisory Panel Pressure Injury Staging System. This change more accurately reflects pressure injuries to both intact and ulcerated skin. - We had planned to undertake subgroup analyses based on type of setting (long-term and acute care) and the type of patient. For this update, four studies were conducted in acute care settings, whilst the others were set in long-term care facilities, and all with geriatric patients. However there was low/moderate heterogeneity between the estimates, thus a subgroup analysis was not undertaken.
Changes in the 2026 update	- We excluded studies where medical-device-related pressure injuries were the primary or secondary outcomes. - We included papers on micromovement, as a positioning regimen, to align with the repositioning descriptors included in the International Clinical Practice Guideline produced in 2025 by the National Pressure Injury Advisory Panel, European Pressure Ulcer Advisory Panel, and Pan Pacific Pressure Injury Alliance [4]. - We added 'category/stage' to describe a pressure injury, in keeping with the various classification systems available worldwide. - For the 11 included studies, we assessed risk of bias (RoB) for the outcomes in the summary of findings tables using the revised Cochrane RoB 2 tool [44] instead of the Cochrane RoB 1 tool [45] that was used in the previous versions of this review. - In keeping with consumer engagement and involvement in this review, we examined patient and public involvement in the included studies using the Guidance for Reporting Involvement of Patients and Public (GRIPP2) checklist [46]. - We modified the wording of the objective.

Table 2. Overview of syntheses and included studies (OSIS) table (clinical trial study characteristics)

Study name (year) country of conduct	Setting	Study de- sign	Intervention	Population (sample size: in- tervention/con- trol)	Outcome do- mains	Outcome measure	Measure- ment time- points	Synthesis method
Bergstrom (2013) USA and Canada	Nursing homes: USA: n = 20; Canada n = 7	RCT	<i>Group A:</i> 2-hourly reposition- ing <i>Group B:</i> 3-hourly reposition- ing <i>Group C:</i> 4-hourly reposition- ing	Nursing home residents aged ≥ 65 years n = 967 Group A: n = 335 Group B: n = 333 Group C: n = 299	Primary out- come: inci- dence of new PI on coc- cyx, sacrum, trochanter, heel	VSA assessment per- formed weekly by li- censed nurses. No information provid- ed on staging of out- come	Weekly fol- low-up over 21-days	1. MA 2. Narrative
Defloor (2005) Bel- gium	32 wards across 11 nursing home	5-arm clus- ter RCT	<i>Group A:</i> 2-hourly reposition- ing with standard mattress <i>Group B:</i> 3-hourly reposition- ing with standard mattress <i>Group C:</i> 4-hourly reposi- tioning with viscoelastic polyurethane mattress <i>Group D:</i> 6-hourly reposi- tioning with viscoelastic polyurethane mattress	Adult nursing home residents n = 262 Group A: n = 65 Group B: n = 65 Group C: n = 67 Group D: n = 65	Primary out- come: inci- dence of new PI, any location	VSA assessment per- formed twice weekly by clinical staff. ACHPR (1992) PI staging criteria fol- lowed	Twice week- ly follow-up over 28- days	1. MA 2. Narrative
Ghezeljeh (2017) Iran	ICU n = 3	3-arm RCT	<i>Group A:</i> 30° HOBE tilt <i>Group B:</i> 45° HOBE tilt <i>Group C (control):</i> routine po- sition	Adults > 18 years n = 120 Group A: n = 40 Group B: n = 40 Group C: n = 40	Primary out- come: unrelat- ed to review (ventilator-as- sociated pneu- monia) Secondary out- come: inci- dence of new PI	2-hourly skin check using study specif- ic Skin Condition As- sessment Form for 3 days. No information on PI staging criteria used	Followed for 3 days	1. Narrative

Table 2. Overview of syntheses and included studies (OSIS) table (clinical trial study characteristics) (Continued)

Jin (2024) China	Intraoperative setting n = 1	RCT	<i>Group A:</i> micromovement protocol intervention <i>Group B (control):</i> silicone dressing (usual care)	Adults > 18 years n = 200 Group A: n = 100 Group B: n = 100	Primary outcome: incidence of new sacrococcygeal intra-operative PI	VSA assessment performed by two nurses. NPUAP (2019) PI staging criteria followed	One skin assessment directly after surgery	1. MA 2. Narrative
Manzano (2014) Spain	ICU n = 2	RCT	<i>Group A:</i> 2-hourly repositioning <i>Group B: (control):</i> 4-hourly repositioning	Adults > 18 years n = 329 Group A: n = 165 Group B: n = 164	Primary outcome: incidence of new PI (at least stage 2), any location	VSA assessment performed by five nurses. EPUAP (1998) PI staging criteria followed	Between study enrolment and ICU discharge. Frequency of assessment unclear.	1. MA 2. Narrative
Moore (2011) Ireland	Long term aged care facilities n = 12	2-arm cluster RCT	<i>Group A:</i> 30° tilt at night with 2 to 3-hourly position changes <i>Group B (control):</i> usual care	Adults ≥ 65 years n = 213 Group A: n = 99 Group B: n = 114	Primary outcome: incidence of new PI	VSA assessment performed at each turning episode by clinical staff and followed up by one researcher. EPUAP (1999) PI staging criteria followed	Weekly follow-up over 4 weeks	1. MA 2. Narrative
Pickham (2018) USA	ICU n = 2	RCT	<i>Group A:</i> real-time wearable patient sensor data with visual reminders for 2-hourly repositioning <i>Group B (control):</i> real-time patient sensor data without visual reminders and with nurse-initiated repositioning (standard care)	Adults > 18 years n = 1312 Group A: n = 659 Group B: n = 653	Primary outcome: incidence of new PI	VSA assessment performed by clinical team and independent wound specialist nurse. NPUAP (2016) PI staging criteria followed	Daily for the first 72 hours	1. Narrative
Tarigan (2021) Indonesia	ICU n = 1	RCT (pilot)	<i>Group A:</i> 2-hourly repositioning <i>Group B:</i> 3-hourly repositioning	Adults > 18 years n = 44 Group A: n = 13 Group B: n = 14	Primary outcome: incidence of new sacral PI	VSA assessment performed by two researchers. NPUAP (2014, 2016) PI staging criteria followed	Daily assessment for max of 14 days	1. MA 2. Narrative

Table 2. Overview of syntheses and included studies (OSIS) table (clinical trial study characteristics) (Continued)

			Group C (control): 4-hourly repositioning	Group C: n = 17				
Young (2004) Wales	Medical ward n = 1	RCT	Group A: repositioning using 30° tilt Group B (control): 90° lateral and supine positions	Adult elderly patients n=46 Group A: n = 23 Group B: n = 23	Primary outcome: incidence of non-blanchable erythema during a 24-hour period	VSA assessment performed by one researcher. EPUAP (1998) PI staging criteria followed	One assessment at 24-hour time point	1. MA 2. Narrative
Zhang (2024) China	Intra-operative setting n = 1	Single-blind RCT	Group A: micromovement protocol Group B (control): silicone dressing (usual care)	Adults > 18 years n = 298 Group A: n = 149 Group B: n = 149	Primary outcome: incidence of new intra-operative PI, any location	VSA assessment performed by one nurse. NPIAP 2019 PI staging criteria followed	One skin assessment directly after surgery	1. MA 2. Narrative
Zhou (2014) China	ICU n = 1	RCT	Group A (intervention): Prone position combined with lung recruitment manoeuvres for 10 hours, alternating 4 to 6 hours supine, and then repositioned in the prone position for another 10 hours Group B (control): supine position with standard care	Adult ICU patients with ARDS n = 116 Group A: n = 56 Group B: n = 60	Primary outcome: unrelated to review (see included studies section) Secondary outcome: incidence of new PI	No information provided on staging of outcome	Daily for up to 72-hours	1. Narrative

AHCPR: Panel for the Prediction and Prevention of Pressure Ulcers in Adults; ARDS: acute respiratory distress syndrome; EPUAP: European Pressure Ulcer Advisory Panel; HOBE: head-of-bed elevation; ICU: intensive care unit; MA: meta-analysis; NPIAP: National Pressure Injury Advisory Panel; NPUAP: National Pressure Ulcer Advisory Panel; PI: pressure injury; RCT: randomised controlled trial; VSA: visual skin assessment